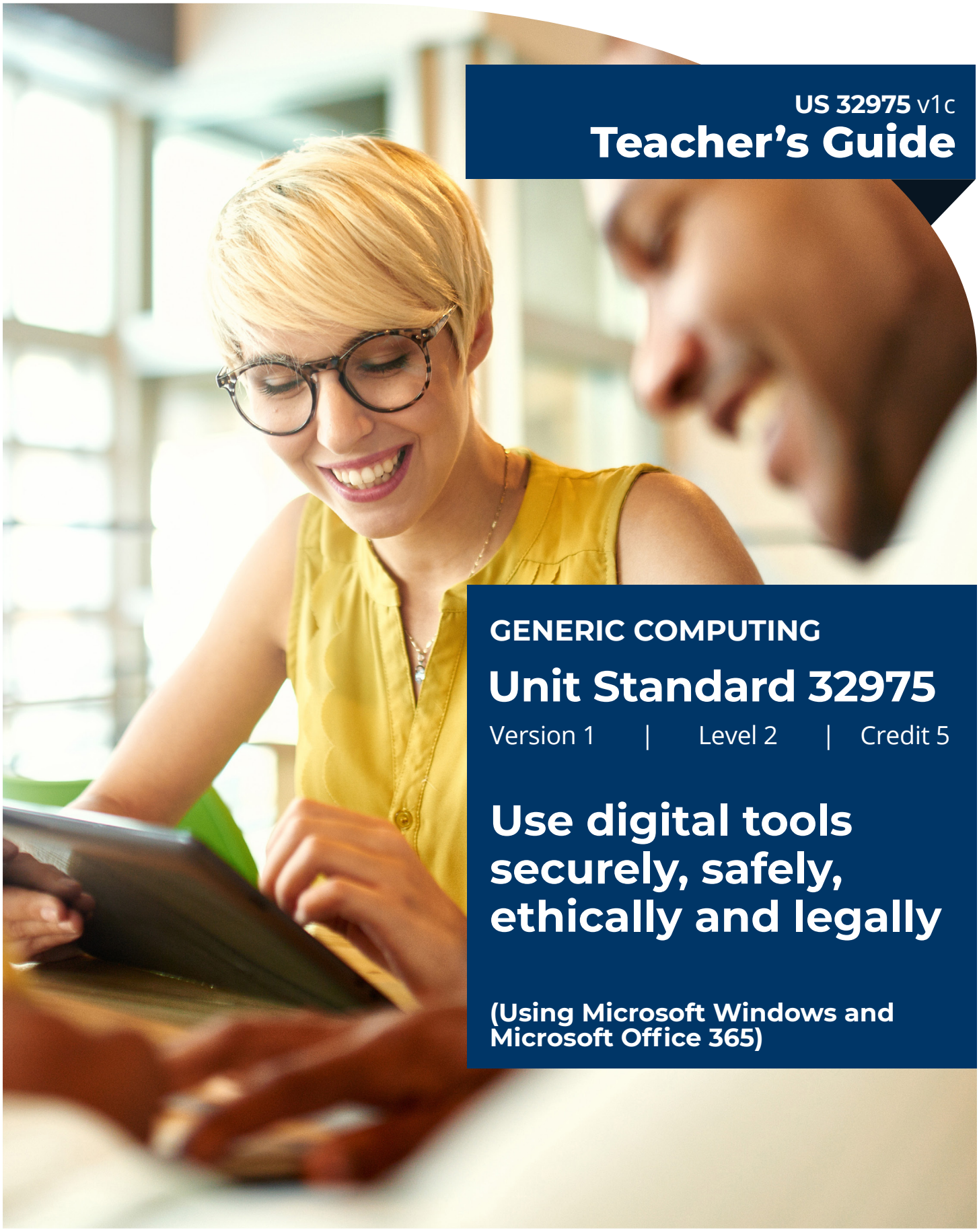


US 32975 v1c

Teacher's Guide

GENERIC COMPUTING

Unit Standard 32975

Version 1 | Level 2 | Credit 5

Use digital tools securely, safely, ethically and legally

(Using Microsoft Windows and Microsoft Office 365)

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NZQA Information

The information contained in this resource is relevant to assessment against the following NZQA unit standard from the Computing – Generic Computing domain.

U/S	LEVEL	CREDIT	TITLE
32975v1	2	5	Use digital tools securely, safely, ethically and legally

The information covered in the resource may also relate to unit standards in other domains.

Level Descriptor: The unit standard is at level 2 of the NZQCF. At this level, the following is expected of the learner:

Knowledge
Has knowledge of some facts and information and understands some concepts relevant to a field of work or study

Practice and performance
Performs a range of familiar activities

Critical thinking
Identifies simple problems or goals and recommends a solution or plan

Collaboration
Contributes to group or team activities and acknowledges views and experiences of others

Communication
Communicates basic knowledge and shares ideas on familiar topics

Environment
Highly structured environments

Autonomy and responsibility
Works or studies under some supervision, with some responsibility for own learning and performance

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About this Teacher's Guide

Learning Purpose & Outcomes

The purpose of this guide is for learners to be able to demonstrate their awareness and knowledge of how to implement basic security when using digital devices and software.

Learners who successfully complete this course will be able to do the following:

- Identify and describe legislation and guidelines relevant to using digital tools.
- Identify risks, procedures and solutions to implement basic security when using tools in a home, work, or study context.

A glossary containing difficult or technical terms has been provided at the end of this guide. These words are highlighted in the main text.

General Information

The teaching and learning guides are designed to prepare learners for the assessment (of the identified NZQA unit standard).

1. This Teacher's Guide is provided for use by teachers only. It is not intended that learners will have access to this document. However, it is expected that, where relevant, teachers will provide learners with suggested answers from this document.
2. These guides are not intended as stand alone distance-learning or self-directed study packages. Teachers should add their own learning resources and activities as appropriate to ensure a robust learning pedagogy. The pedagogy engaged should be suitable to the particular organisation or school the teacher is employed in. This includes appropriate extension learning as required.
3. Learners may need additional literacy and numeracy support as they work through this course. The teacher needs to ensure that this support is in place for students.
4. Website and application (including apps) content changes regularly. This may mean that changes have been made to content that is mentioned in the Learner's Guide and the Teacher's Guide. It is important that any links or applications are previewed to ensure they are active or current, before giving them to students.
5. Laws and compliance regulations are regularly amended and/or repealed. Teachers should research and incorporate any recent legislative or compliance changes where required.

Teacher's Note:

This symbol includes notes for teachers. It is included throughout this guide to assist teachers to deliver learning content.

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LESSON 1:

Requirements and responsibilities when using digital tools

Learning Objectives

When you have finished this lesson, you will be able to identify and describe the following legislation and guidelines for the use of digital tools:

- Information privacy
- Copyright
- Software licensing

A space has been left on the right of every page for you to make notes about what you are learning.

Introduction

In this course, you will learn about the ethical and legal requirements that are relevant to the use of digital devices and software. You will also learn about the risks you could face when using digital tools, and how to implement basic security to manage these risks.

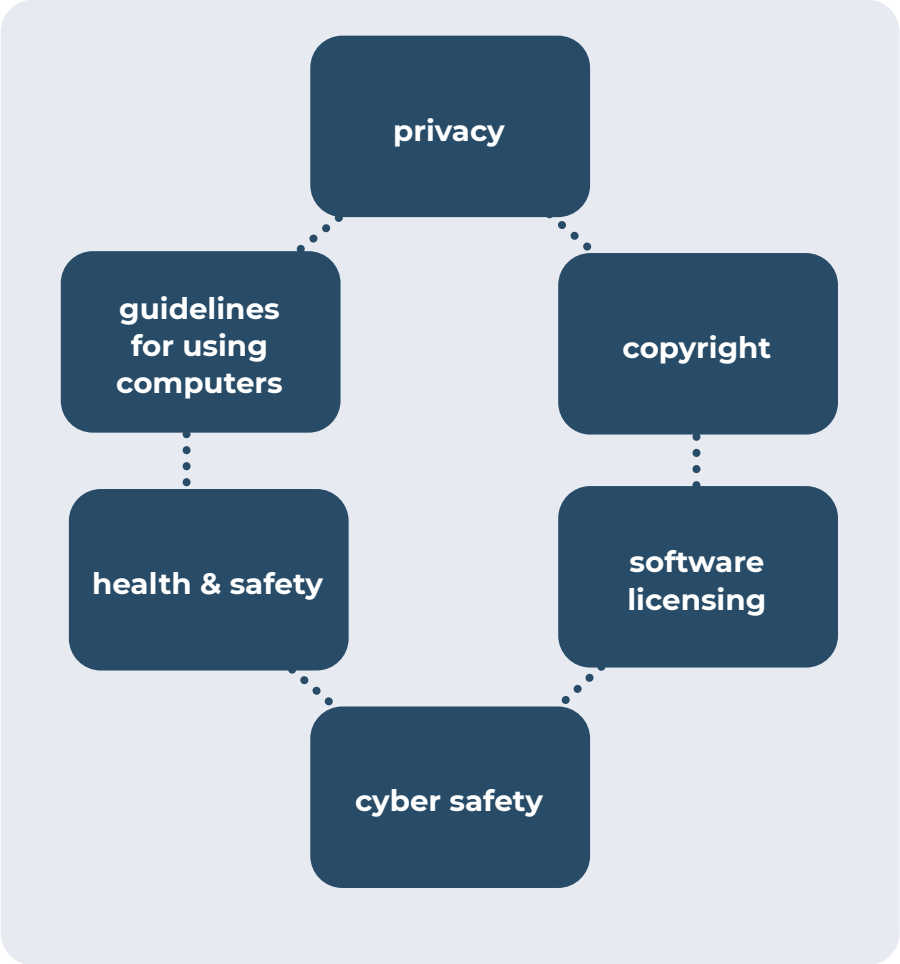
ethical	following morals and principles - displaying respectful and responsible behaviour.
legal	behaviour that is required to comply with the law

Many legal requirements are also ethical ones - often the law requires people to behave ethically. Sometimes however you can follow the law but still display unethical behaviour.

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LESSON 1: **Requirements and responsibilities when using digital tools**

In Lessons 1 and 2, we will look at some key types of legislation (laws) and guidelines that apply to the use of digital tools. This includes the following:



These laws and guidelines all aim to protect you and others when digital tools are used. Following all legislation will also help you avoid any punishment under the law (eg a fine, etc). Let's look at these laws and guidelines, and how they relate to the use of digital tools.

Information privacy

Organisations hold personal information about individuals, eg staff home address, telephone numbers, customer details, etc

compliance requirement	A law, rule, or standard that must be met.
personal information	Anything that can identify a person. For example, a photograph of the person, their name, address, etc.

Under New Zealand law, everyone has the right to privacy of their personal information.

The Privacy Act 2020 sets rules for when and why personal information about individuals can be collected and released.

- It also includes rules for how and where personal information should be stored.
- The aim of the law is to give individuals greater control over their personal information.

The Privacy Act lists thirteen **Information Privacy Principles**. These principles deal with how personal information can be collected and used. A summary of the principles is given in the information box on the next page.

- If you work for an organisation, it is important that you are aware of the privacy principles, and that you take all reasonable steps to treat any personal information you have access to as private.
- If you give your personal information to any organisation, it is also important that you are aware of the privacy principles and the control the law gives you over your information.

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LESSON 1: **Requirements and responsibilities when using digital tools**

Information Privacy Principles

1. Collection of information must be for a lawful reason.
2. If possible, you should collect the information directly from the individual person.
3. The individual should be made aware that their information is being collected.
4. The information should not be collected in an intrusive manner.
5. The information collected must be securely stored.
6. The person has the right to access information you have about them.
7. The person has the right to correct information you have about them.
8. You need to take all reasonable steps to ensure the information is accurate.
9. Personal information should not be kept longer than needed (for the purpose it was collected).
10. Information obtained for one purpose should not be used for another purpose without the consent of the individual.
11. Other people should not be given access to the personal information without good reason.
12. There are strict guidelines to be followed when sharing personal information outside New Zealand.
13. A unique identifier (instead of the person's name) should only be used if needed.

What do You Think?

If you send an email to several recipients, with all recipients' email addresses in the **To:** field, you may be revealing personal information.

What personal information are you revealing?

People's email addresses (and possibly also their names).

Why could this be a problem?

Someone else could use these email addresses to send people unsolicited information (eg advertisements), or to contact them.

LESSON 1: **Requirements and responsibilities when using digital tools**

Being ethical

Not only is it a legal requirement to treat personal information with care, it is also an ethical one. Many principles of the Privacy Act also encourage ethical behaviour. Remember, being ethical is following principles. For example:

- It is not respectful to discuss someone's personal business in a public space where it could be overheard, or with someone who is not allowed to know the information.
- It is not ethical to collect more information than is needed.
- Even if you know the person, it is not ethical to look up their personal information just because you're curious.

Sometimes even though the behaviour is strictly legal, it may not be ethical.

Example

JK Financial Services collects customers' personal information when customers use their services. Its privacy notice clearly gives the client the option to opt in to hear about special offers. They use this data to target customers with advertising that exploits them. An example of this is marketing high interest loans to customers it knows are in financial difficulty and need money quickly.

The above example takes advantage of customers and does not treat them fairly and respectfully.

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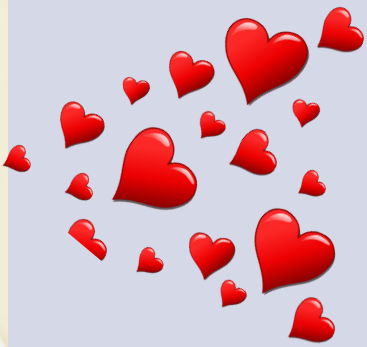
Try it for Yourself

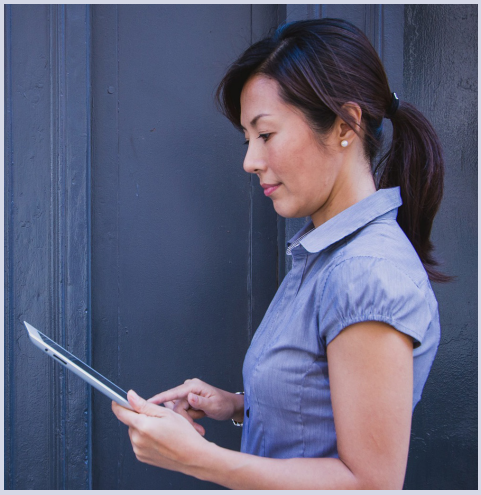
Read the scenario below. Then answer the questions that follow.

As part of his job, Tana is responsible for paying staff wages. He has access to all staff members' personal information, such as their email address, phone numbers, and home addresses. Two staff members at Tana's work are Mark and Mandy. Mark really likes Mandy and they have dated once but it is not serious. Mark asks Tana if he can tell him what Mandy's home address is, so that he can surprise her with some flowers on Valentine's Day.



Mark





Mandy

1. Mark seems like a nice guy. Is it okay for Tana to give him Mandy's address so he can surprise her on Valentine's Day? Give a reason for your answer by referring to the relevant Privacy Principle(s).

It is definitely not okay for Tana to give Mandy's address to Mark. Under Privacy Principle 10, you are not allowed to use information gathered for one purpose (eg paying wages, etc) for another purpose without getting permission from the person. Also, according to Privacy Principle 11, you must not share someone's personal information with anyone else without good reason.

Continued on next page...

LESSON 1: **Requirements and responsibilities when using digital tools**

2. One day, while Tana is working in the database containing staff details, Mark comes into his office for a chat. During their chat, Tana is called out of the office, but Mark stays behind, saying he will wait for Mark to come back. Tana rushes out the office while he is still logged into the staff database. He does not log off from his computer. Which Privacy Principle(s) is Tana not adhering to and why?

Principle 5 – because he is not dealing with the information securely. By leaving the room while he is still logged into his computer and the database program, there is the potential that Mark (or someone else) could access the staff members' personal information. For this reason, he is also not adhering to Principle 11, because he could potentially be allowing Mark to access the personal information.

3. Tana has moved to another position and Mark now has Tana's role. Would it be ethical for Mark to access the system to get Mandy's address to send flowers?

No, it would not be ethical, it is not a moral or responsible use of the information. It would also be illegal as Mark would not be accessing the information for the purpose it was collected for.

Unsolicited Electronic Messages Act 2007

Spam is any unsolicited electronic message (ie a message you did not ask for). The **Unsolicited Electronic Messages Act 2007** prohibits spam being sent to, from, or within New Zealand. This includes email, instant messaging, SMS and mobile phone messaging but NOT voice or fax messages.

Sending unsolicited messages commercially is an example of using personal information for a purpose other than that it was collected for.

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Copyright

Copyright Act 1994

The Copyright Act 1994 sets out the copyright restrictions in New Zealand. Under the Act, any original work created, published and/or performed in New Zealand is automatically subject to copyright.

Items subject to copyright often display the copyright symbol ©, the name of the copyright owner and the date.

However, in New Zealand an original work is still copyright-protected, even if it does not display the copyright symbol.

	**Copyright** literally means 'the right to copy'. So, the person (or organisation) which holds the copyright for a particular item has the following exclusive rights. • To copy, publish, distribute, perform, broadcast, or adapt their work. • To give up copyright on their work. Below are some examples for how a person might give up copyright to their work
	**Creative Commons** is where an author of some content (such as a graphic) allows it to be used under certain conditions without the user needing to pay for it. A list of the types of Creative Commons licences in New Zealand can be found at: https://creativecommons.org/about/cclicenses/
	**Public Domain** is a special type of Creative Commons licence which allows you to use images for **any purpose** without needing to acknowledge the source of the image.

In most cases the person who created the work is also the owner of copyright to that work.

There are two exceptions:

- When an employee creates a work for their employer, then copyright belongs to the employer.
- When a particular work is commissioned by someone who pays for it. Then the copyright belongs to the person who commissioned the work.

For example, copyright of this learning guide belongs to Instant Education Solutions Ltd. It does not belong to the person who actually wrote the guide.

There are some cases where 'special permissions' to use copyright material is given under law. These allow greater copyright permissions than normally allowed under the law such as a single copy by a student or researcher for their personal use or an education licence.

Copyright (New Technologies) Amendment Act 2008

The Copyright (New Technologies) Amendment Act 2008 amended the Copyright Act 1994 to make sure that it is relevant for today. Technology has changed drastically since the Copyright Act came into force (in 1994). The Act clarifies the definition of 'copying' and 'copy' so that it clearly includes making digital copies of works.

Another change to the law is that it clarifies that digital copies made for educational purposes can be communicated to 'authenticated users' (eg students and staff). This means that an educational institution can store a copy of work available on a website if it is used for an educational purpose (eg for a course).



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Ethical treatment of copyright works

Copyright law largely enforces ethical treatment of a copyright owner's work. For example, not only is it illegal to pass someone's work off as your own, it is also unethical. There are also other instances where ethical behaviour is also important, regardless of copyright law.

Examples include:

- Acknowledging the work's creator, even if you don't have to. If the image is in the public domain, you may still want to acknowledge the original photographer or artist.
- Considering images that show people's faces - are they comfortable with the use of the image?

It is unethical to use the work in a way that changes its original meaning or makes it seem that the creator agrees with or supports your message.

Try it for Yourself

Read each of the case studies that follow. Each of these people work for a company called Phob International. For each say whether the people are complying with copyright legislation. Give a reason for your answer.

1. Katie needs to make copies of the Phob International brochure to send to clients.

This is fine because the company (Phob International) owns the copyright to the brochure.

Continued on next page...

LESSON 1: **Requirements and responsibilities when using digital tools**

2. Aroha wants to copy all the invoices received over the last year and send them to the accountant.

This is fine. The invoices were created by the company and are not subject to anyone else's copyright.

3. Ben heads up the sales department. He recently bought a very useful book on sales techniques that he wants to share with someone else in his department. He makes a single copy of the entire book.

This is a breach of copyright because he has copied an entire book he does not own the copyright to. (He only owns the book, not the copyright to the book.)

4. Shelley needs an image to use on the company's website. She finds the perfect image on the internet, and uses it on the site, without checking the copyright licence.

Shelley may be breaching copyright because the image may not have been released into the Public Domain. She must check and comply with the licence.

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Software licensing

Software, like any other item created or developed, is subject to copyright. You may be breaking copyright law if you copy or use software without a licence.

You can get a licence in several different ways, including:

- paying for the software
- downloading software that is clearly indicated as being provided for free
- downloading trial software (which is generally provided for free).

You are buying or downloading a **licence to use** the software. You don't actually own it. Any use outside the terms of the licence may be a breach of copyright. Therefore you cannot copy it or change the code.

The exception is if the software is being provided under a licence that allows you to alter, use, copy, and/or distribute the software. Open source software is an example of this.

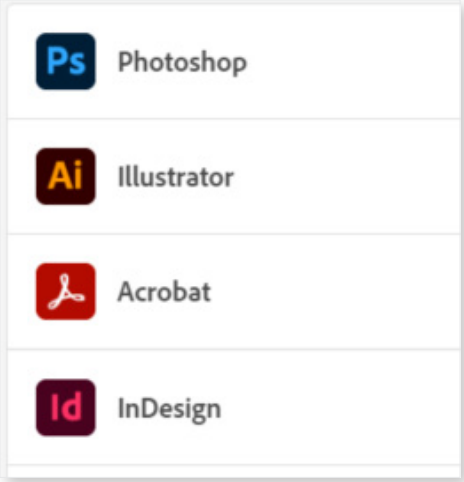
Open source software	Software provided for use, modification, and redistribution. It can usually be downloaded from the internet at no/little cost. Examples include Gimp & Inkscape for graphics, Scribus for desktop publishing and SwatchBooker.

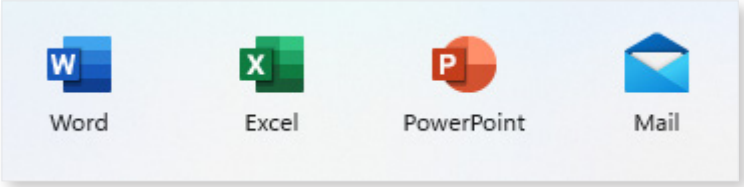
LESSON 1: **Requirements and responsibilities when using digital tools**

Increasingly, software is being made available on a software as a service (SaaS) basis.

With SaaS, you could think of the software as being 'rented' to the organisation or individual. The organisation/individual will only have access to the software if they continue to pay the subscription fee.

Software as a service	A type of software licence where the software is licensed on a subscription basis. Usually the software is paid for on a monthly or annual basis. Examples include Microsoft Office 365, Dropbox, Google Apps, Adobe Creative Cloud





It is common for people to simply say they have read and/or accept the terms of a licence agreement, without actually reading them. This can lead to someone unintentionally breaking the terms of the agreement. However, they (and the organisation they work for) could still receive a severe penalty if they break any aspects of the licence agreement.

There are two key ways in which licence agreements could be broken.

- **Knowingly and deliberately** using pirated software and/or sharing it with others.
- **Using the software outside of the licence terms.** For example, this could include licence terms related to:
  - the number of devices on which the software may be installed
  - whether it can be used for commercial purposes (or only for personal use)

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Behaving ethically

Ethical treatment of the software licence is following its terms of use. As an example, if the organisation buys a licence for one person to use it, they should not then make multiple copies or install it on multiple machines for others to use.

What do You Think?

1. How many times have you simply selected 'I accept the terms of the Licence Agreement' without first reading through the agreement carefully?

2. An organisation has purchased a single licence for online software. This gives them one login to use. Is it ethical to share that login amongst staff?

No. It isn't legal either. The licence is for one user, not multiple.

LESSON 2:

Health and Safety

Learning Objectives

When you have finished this lesson, you will understand the legal requirements for acting in a healthy and safe manner when using digital devices. This includes the following legislation:

- Health and Safety at Work Act (HSWA) 2015
- Harmful Digital Communications Act 2015

Health and Safety at Work Act 2015

The Health and Safety at Work Act (HSWA) 2015 sets out the requirements, duties, and rights related to workplace health and safety. The aim of the Act is to help ensure the health and safety of workers and others (eg customers, clients, visitors, etc), in a workplace.

It outlines specific duties that both organisations and workers have to ensure a safe and healthy working environment.

Organisations are required to do the following:

Provide and maintain a working environment that is without risks to health and safety. In the context of digital work (eg working with digital tools and information), this includes risks related to the:

- physical environment (eg equipment and how it is positioned, lighting, ventilation, dust, heat, noise, and risk of physical injury or discomfort)
- psychological environment (eg work-related stress and tiredness).



Provide workers with information, training and/or supervision to ensure they are able to protect themselves (and others) from health and safety risks related to working for the organisation.

Monitor the health of workers and the working environment to prevent future injury or illness.

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In the context of digital work, this could include:

- ergonomic risks such as strain, fatigue, or pain from repetitive motions, poor lighting, etc
- physical risks such as noise, heat, exposure to chemicals (eg from a printer).

Workers are required to do the following:

- Take reasonable steps to care for their own health and safety.
- Take reasonable steps to avoid negatively impacting the health and safety of others (eg other workers, customers, clients, visitors, etc).
- Follow any reasonable health and safety policies and procedures their organisation has in place, and which they have been informed about.
- As far as reasonably possible, follow instructions given by an organisation that will enable the organisation to meet the HSWA requirements.
- Where the worker is unable to take steps themselves to deal with something negatively affecting health and safety, they need to make sure their employer is made aware of the problem.

As you can see from the above, it is not just the responsibility of an organisation to avoid and manage risks to health and safety. Individual workers also have important duties in this regard.

In this lesson, we will focus on the health risks and safety issues most common to people working or studying in a digital environment. **NOTE:** Wherever you see reference to a 'worker' in the following sections, this can also refer to someone studying in a place of learning.

Risks related to the work environment

The lighting, noise levels, temperature and other aspects of the working environment can lead to physical discomfort or fatigue for workers. We will look at this in the next section.

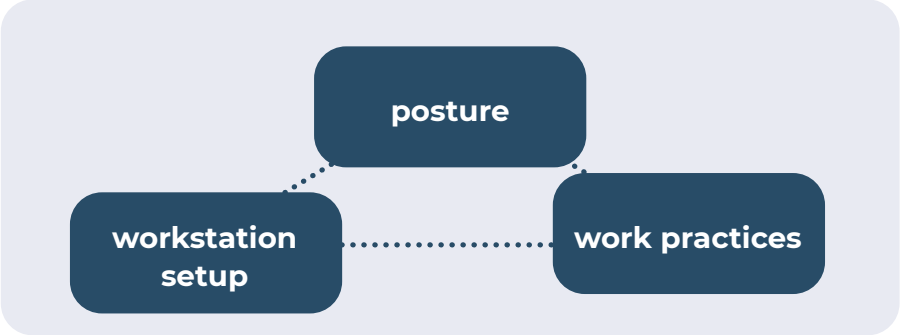


Guidelines for using computers

Teacher's Note:
The ACC Guidelines for Using Computers is no longer available. This section refers to both WorkSafe guidelines and general safe use of computers.

The Health and Safety at Work Act (2015) deals with health and safety in the workplace, which includes the use of digital tools. WorkSafe have provided guidelines for using computers which can be found at https://www.worksafe.govt.nz/topic-and-industry/work-related-health/musculoskeletal-disorders/working-with-computers/.

Ares you need to consider include:

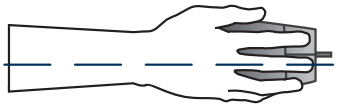


Posture

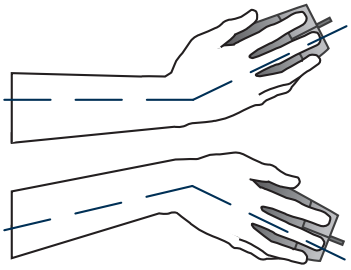
Posture is the way in which you position your body.

Having a good posture when using a computer can help reduce the chances of you suffering from strain, pain, or injury.

A **neutral body position** helps to reduce stress and strain on your muscles and bones. It is a comfortable working posture where your joints are lined up in a natural way.



1. Fingers, wrist and forearm lined up in a natural, comfortable way.

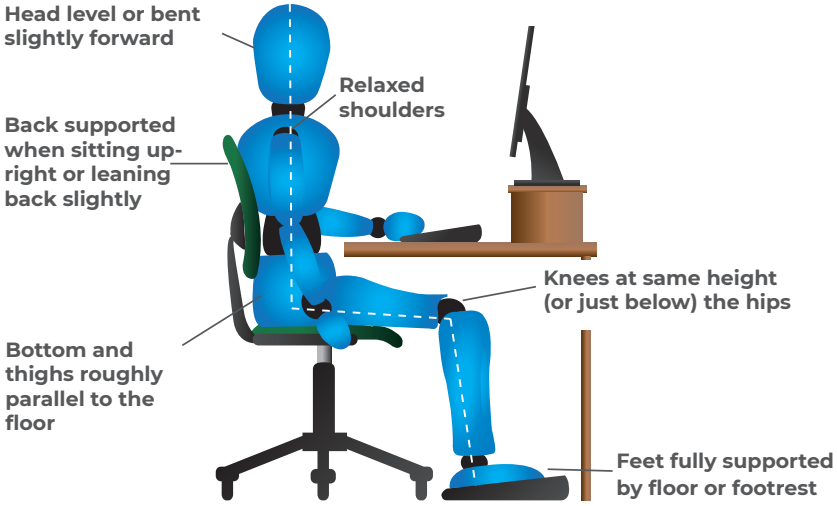


2. Wrist at an awkward angle to the forearm. This can cause stress and strain on the wrist and arm.

When using a computer you change your posture regularly. For example, you should not sit in the same position (even if it is a neutral position), for too long. Having the same position for too long can lead to stress and strain on your body.

The diagram that follows summarises key points related to a correct posture when working at a computer.

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Healthy work practices

An important aspect of safe work practices in a digital environment is to take regular rest periods and exercise as part of your standard work practice. This helps to guard against physical discomfort and strain, physical and mental fatigue, and stress.

Workstation setup

In the section on Health and Safety we learned that the furniture, equipment, and computer components that are selected, and how they are arranged, can have an impact on the physical wellbeing of a computer user. The diagram below, illustrates the key items that make up a typical computer workstation.



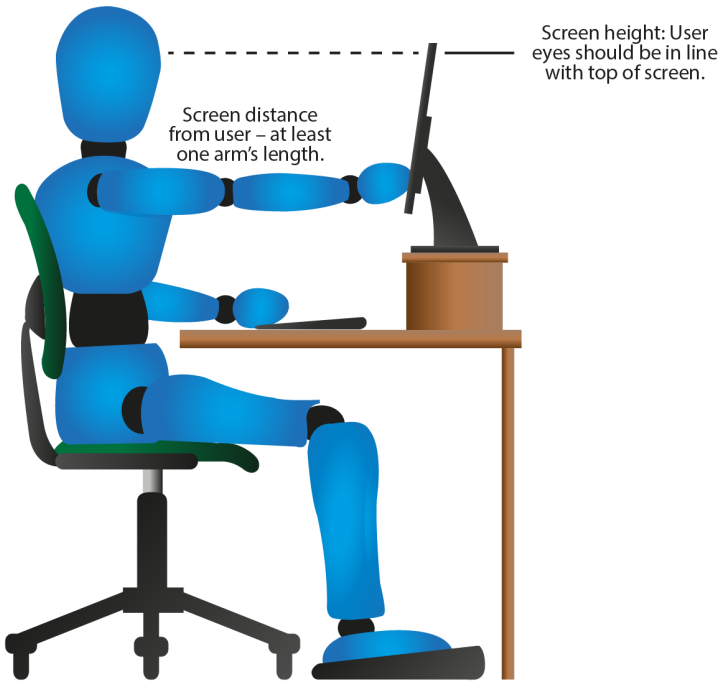
LESSON 2: **Health and Safety**

Your workstation should be set up to:

- allow the user to work comfortably without having to stretch or position themselves or parts of their body (eg wrist) in an awkward way
- enable the user to move easily and change their body position regularly
- allow the user to do several different types of tasks - eg looking at things on the screen, typing on the keyboard, using the mouse, writing, and reading.
- can be adjusted to cater for people of different heights, etc.

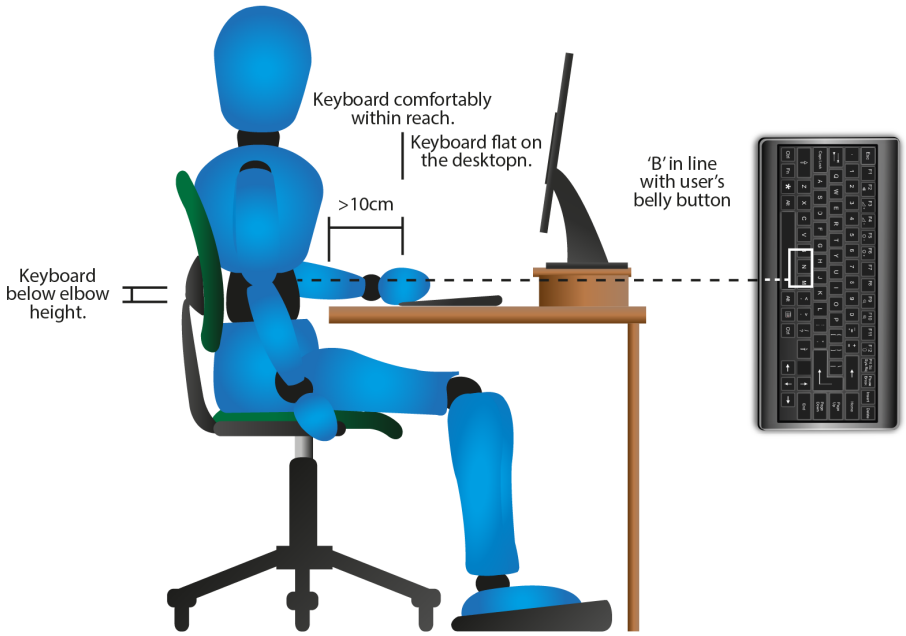
The diagrams below summarise some of the key points related to the arrangement of workstation elements.

Positioning of your monitor/screen

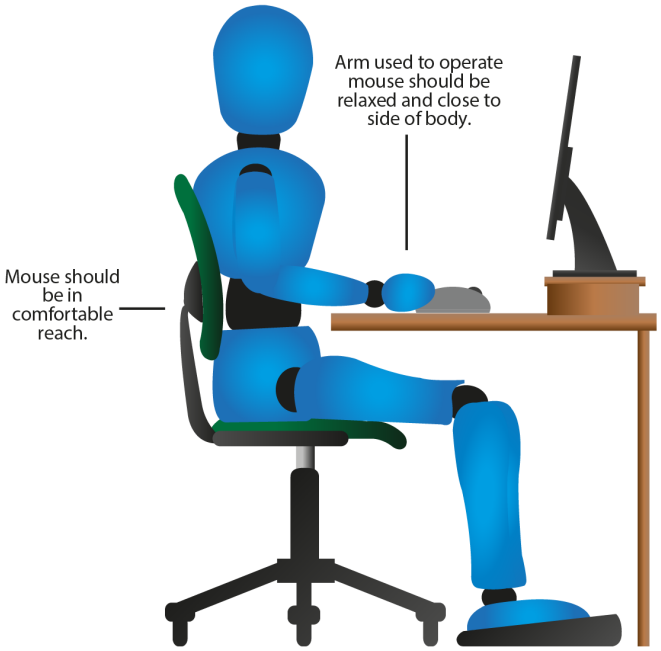


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Positioning of your keyboard



Positioning of your mouse



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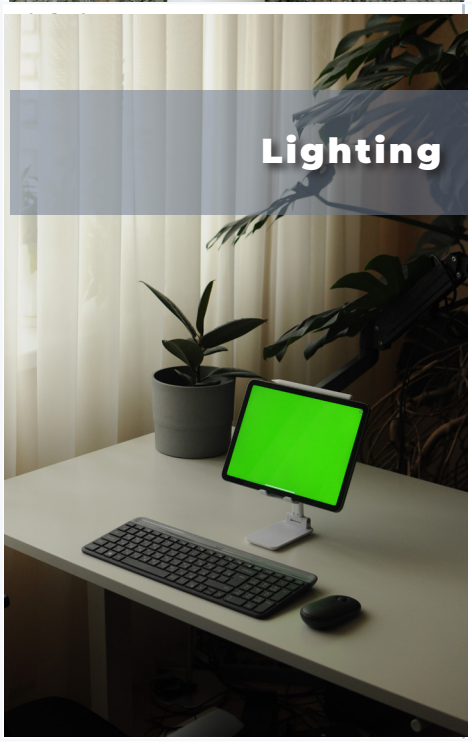
Along with the set-up of the equipment and physical work practices, the environment of your workplace can also affect your health in the workplace:



- Personal space is important so people don't feel 'crowded'.
- People need to have enough space to be able to do their work safely and comfortably.
- There needs to be enough space for a person to have a range of comfortable working postures.
- The amount of space needed will depend on the individual (eg their height), their type of work, and hardware being used.



- Floors need to be kept tidy.
- Access ways need to be kept clear, so that users can get to/ from their workstations easily, without anything blocking their way.
- Arrange power and other cables so they are not a tripping hazard.



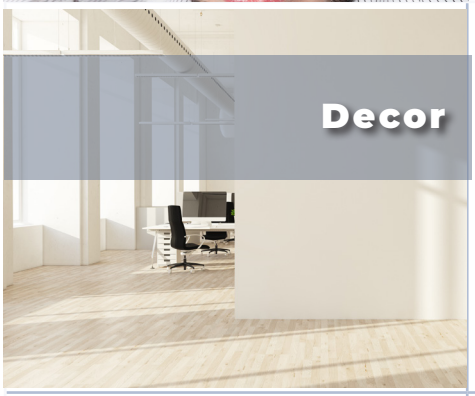
- Poor lighting (ie not light enough, or too bright): This can lead to sore and strained eyes, headaches and even migraines.
- Lighting which causes glare or shadows. For example, if there is light shining directly on your computer screen, this can make it difficult for you to see what is on the screen. As a result, you may strain your eyes, or lean forward to try to see what is on the screen. This could result in pain and discomfort.
- If there is a big contrast (difference) in the brightness of the screen and the surrounding environment, this can cause eye strain. For example, if you are working in a dark room, then the screen could be too bright for your eyes. You might need to adjust the screen brightness or add lighting in the room.
- One way to avoid these problems is to position the workstation correctly in relation to light sources.

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- Noise can make it hard for people to communicate and/or concentrate.
- High levels of noise can lead to headaches and other physical discomfort and can be tiring.
- Noise levels should be as low as possible.



- Décor affects the lighting in a room and so should be planned along with the lighting.
- Generally, ceilings should be brighter than walls; walls should be brighter than floors.
- Avoid sharp contrasts in brightness (eg a white ceiling and very dark wall).



- The room should be well-ventilated. This is to help ensure that any airborne pollution is removed.
- The room should be heated to a comfortable temperature.



- There are several potential hazards associated with printers, including noise and fumes.
- To avoid these hazards the printer should be far enough away from users that they are not affected.
- A good air filtration system will help to remove any fumes created by the printer.
- If possible, a printer should ideally be in a different room to the users.

Cyber safety

With so much communication online, it is important that we:

- act in a safe manner toward others when communicating
- keep ourselves and our personal details safe.

Acting in a safe manner towards others

Not only is it good manners to be polite and respectful to others, there is also legislation surrounding how you communicate online. The **Harmful Digital Communications Act** governs the content of online communications.

The purpose of this law is to prevent harm caused to people by digital communications, and to provide victims of harmful digital communications with support. Examples of harmful communication include cyberbullying (eg bullying someone on social media) or cyberstalking (following a person online, tracking their comments/movements etc).

Other types of communication content the law covers include making false allegations or making hurtful or harmful comments about someone because of who they are eg their race, colour, gender, sexual orientation or disability.



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Keeping ourselves and our information safe

People share a lot of personal information about themselves online. They also make perform financial functions online, such as bank transactions and online payments.

Ways that you can keep yourself and your personal information safe include:

- Use security such as passwords to access your device and apps.
- Many apps insist on passwords, you should never divulge these. You will look at this in more detail in Lesson 3.
- DO NOT share personal information on social media such as your date of birth, address, phone numbers and email addresses, licence or passport numbers, or anything else that might identify you.

Not only could this information be used to commit identity theft eg someone could pretend to be you and access credit in your name, it could also be used by someone such as a stalker to find you.



Deciding which laws apply in specific situations

Different laws may apply to different situations or particular people. For example, a person who works as a sales person will be dealing with customers' personal information. They will need to take the Privacy Act 2020 into account. However, a student may not need to take this law into account because they will generally not be dealing with the personal information of other people.

Look at the following examples:



Name: Marcus

Position: Team Leader at IT Wizards

Marcus needs to be aware of the following laws as part of his work.

Copyright Act 1994 – Marcus needs to make sure he does not copy any documents or other resources (eg images) in a way that goes against the terms of the Copyright Act.

Copyright (New Technologies) Amendment Act 2008 – Marcus must not download any copyrighted media (such as movies or music). His workplace could be fined or prosecuted if he does this.

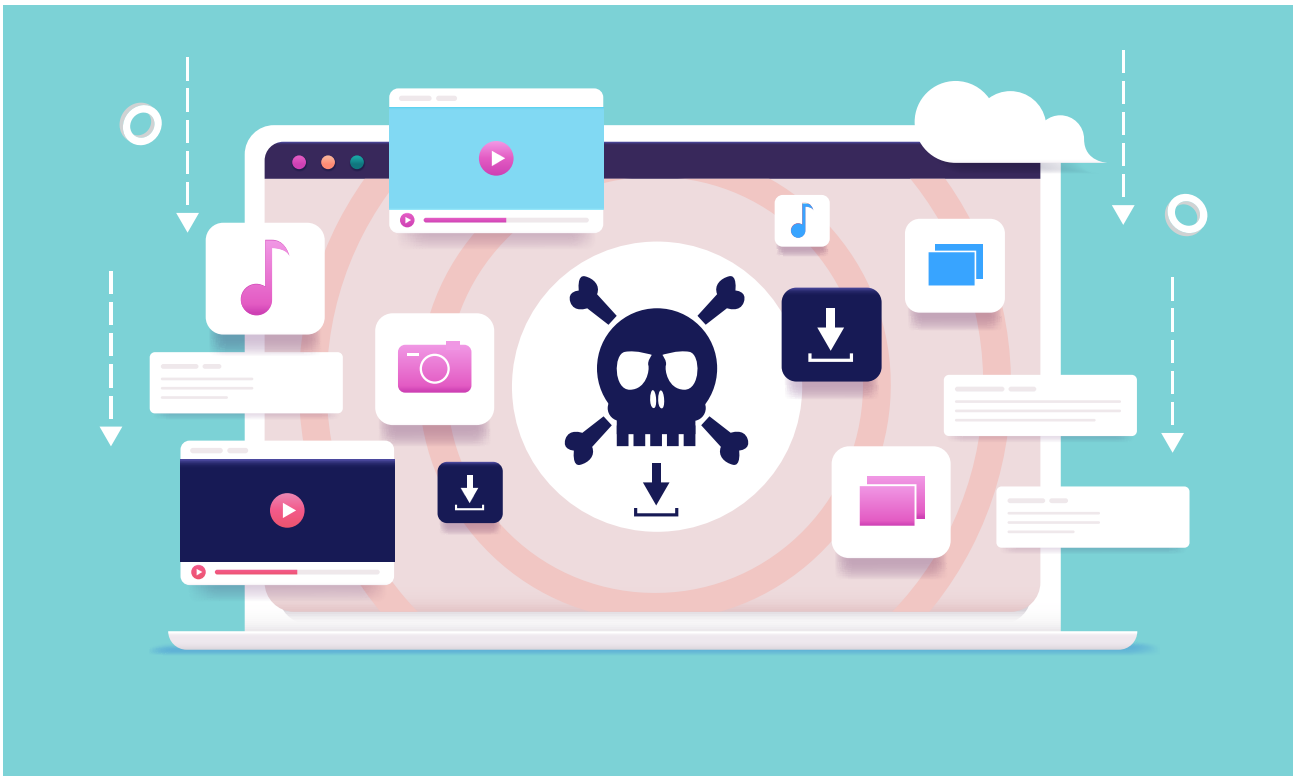
Harmful Digital Communications Act 2015 – Marcus must not send any messages that could be harmful or abusive to clients or colleagues. He could be prosecuted for doing so.

Privacy Act 2020 – Marcus must make sure that all customer information is stored securely and only used for the purpose specified to the customer when it was collected.



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Name: Samantha

Position: Student at Fairview High School

Copyright Act 1994 – Samantha must make sure any material she uses complies with the terms of the Copyright Act. There are specific exemptions in the Act that allow people to use certain resources for education use.

Copyright (New Technologies) Amendment Act 2008 – Samantha must not download any media (films, music etc) illegally.

Harmful Digital Communications Act 2015 – Samantha must not send any messages that could be harmful or abusive to teachers, classmates, or people outside the school. She could be prosecuted for doing so.

Review

Read each of the case studies below, and then do the following for each.

a. Identify the type of legislation/guidelines that would be relevant to the situation. Choose from:
   - guidelines for using computers
   - information privacy
   - copyright
   - health and safety
   - software licensing.

b. Describe the legislation/guidelines that would apply to the situation. You need to include the name of the specific legislation/guidelines that would apply and give examples to support your descriptions of the legislation/guidelines.

1. Matt works as a software developer for a small company. He often works up to 10 hours a day, on a computer. Much of the work he does is repetitive such as typing, clicking, and scrolling on his mouse, etc.

   a. Which type of guidelines are relevant to Matt's working situation?

Guidelines for using computers

   b. Describe the guidelines relevant to Matt's working situation. Remember to include the name of the specific guidelines that may apply, and give examples to support your description.

The ACC Guidelines for Using Computers – Preventing and managing discomfort, pain and injury *will apply in this situation because they outline guidelines for working safely when using a computer. Because Matt works long hours at the computer, it is important that he:*

- *adjusts his chair, desk, and computer components (eg mouse, keyboard), so that he can work comfortably, in a neutral position, without needing to strain*
- *positions items to allow for a comfortable working position without strain – eg positioning his monitor to make sure there is no glare on the screen or light shining directly in his eyes*
- *takes regular breaks*
- *positions his body (posture) in a neutral position, and changes his position regularly throughout the day*

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LESSON 2: **Health and Safety**

2. Mina works as a sales manager at a large company. One of her clients asks her to email him the personal cell phone number of another client, because he wants to call him about something urgent. Mina knows both the clients well, and knows that they are both good, trustworthy people.

   a. What is the legislation that would apply?

Privacy Act 2020

   b. Describe the legislation relevant to this situation. Remember to give examples to support your description.

The Privacy Act outlines the rules relating to how personal information may be collected, how it must be stored, and how it can be shared. In this case, Mina needs to keep in mind two key principles of the Act: that she can only use information for the purposes for which it was collected (Principle 10) and that she cannot share the information with someone else without good reason (Principle 11).

Continued on next page...

LESSON 2: **Health and Safety**

3. Sam is completing a project with two other classmates. The due date is tomorrow and they haven't even started yet. However, they have found one good source of information online that has all the details they need. They are considering copying part of this source to use in their project, without referencing it. They think this is the only way that they will finish their project on time.

   a. What is the legislation that would apply to this situation?

The Copyright Act 1994 and Copyright (New Technologies) Amendment Act 2008

   b. Describe the legislation relevant to this situation. Remember to give examples to support your description.

These laws set out the restrictions that apply to copyright in New Zealand. According to the law, any original work is automatically subject to copyright – including anything on the internet. Sam and her team members must be careful not to copy too much of the text, and for the parts they do include, they need to include a reference which indicates who originally created the text.

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LESSON 2: **Health and Safety**

4. Nico works in the administration department of a large law firm. He often works very long hours and is responsible for communicating with clients, lawyers in the law firm and from outside the law firm, etc. His work can get quite stressful because he sometimes needs to deal with clients (and lawyers) who are not very polite and/or are angry, etc.

   a. What is the legislation that would apply to this situation?

Health and Safety at Work Act 2015

   b. Describe the legislation relevant to this situation. Remember to give examples to support your description.

This act sets out the requirements, duties, and rights of both organisations and employees to ensure a safe and healthy workplace environment. This includes both physical and psychological safety and health. In this particular situation, Nico needs to inform his employer if he is starting to feel the effects of stress, and he should also take steps to manage his own stress (eg by taking regular breaks). He also needs to make sure he doesn't contribute to anyone else's stress by, for example, behaving in an aggressive way towards clients/lawyers. He also needs to do what he can to make sure his working environment will not harm his physical health (eg making sure nothing is left on the floor that could be a tripping hazard). Where there is something that could be negatively affecting his health or safety, but he cannot do anything about it himself, he should make his employer aware of the problem.

Teacher's Note:

Learners may also refer to the **Harmful Digital Communications Act** (as it pertains to cyberbullying and other harmful communications), as well as the **Unsolicited Electronic Messages Act** (as it pertains to sending electronic messages to clients who have not asked for that communication).

Continued on next page...

LESSON 2: **Health and Safety**

5. Phil needs to create a digital presentation for a class project. However, he does not have any digital presentation software. He decides to go online to see if he can find software to download.

   a. What does he need to check to make sure he is allowed to download the software?

The software's licence

   b. Describe the legislation relevant to this situation. Remember to include the name of the specific legislation that may apply, and give examples to support your description.

Software, like any other original work, is subject to copyright (so the Copyright Act may apply). When you buy software, you do not buy and own the software iself, you only own the licence to use the software. The licence will set out the conditions under which you can use the software. Phil needs to make sure that the software he downloads is legitimate (ie sourced directly from the creator/distributor's site), and that he uses the software in keeping with the licence. This could include, for example, checking how many devices he is allowed to install the software on, and what it may be used for.

6.

a. Tessa has signed up for a new social media account. What personal information should she mark as private?

Address

Date of birth

Mobile

Email

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LESSON 2: **Health and Safety**

6.

b. Tessa is very proud of herself as she has just achieved her Learner's Drivers licence. She wants to tell her friends by putting a picture of it on her social media page. Should she do that? Why/why not?

No, because someone may use it to commit identity theft.

c. Tessa has to be careful about what she says to others online. What legislation governs online communication?

Harmful Digital Communications Act.

d. What are some examples of comments Tessa need to avoid making online?

Bullying, putting others down because of their gender, race, religion, or disabilities.

LESSON 2: **Health and Safety**

Review sheet

Consideration	Legislation/Guideline	Example
Information privacy	Privacy Act 2020 Unsolicited Electronic Messages Act 2007	When personal information is collected about individuals eg customers or potential customers, there are ethical and legal considerations. For example: • the information cannot be used for a purpose other than the reason it was collected for • the individual must be aware it was collected and give their permission • it cannot be shared without good reason.
Copyright	Copyright Act 1994 Copyright (New Technologies) Amendment Act 2008	The person or organisation who created a work, be it written, audio or visual, automatically owns its copyright. Permission must be granted to use the work except for specific circumstances.
Health and safety	Health and Safety at Work Act 2015 ACC Guidelines for using Computers Harmful Digital Communications Act 2015	Both employers and employees have a responsibility to keep themselves and others safe in the workplace. When using digital tools, this includes following the ACC Guidelines for Using Computers. Health and Safety considerations include • workload • posture • workstation layup and location • physical environment.
Software licencing	Copyright Act 1994 Copyright (New Technologies) Amendment Act 2008	Purchasing software doesn't mean you own it, you own the licence to use it. Unless the licence allows, the software cannot be copied or altered in any way
Cyber safety	Harmful Digital Communications Act 2015	When communicating with others online you must ensure that you are polite and respectful at all times. You must not say anything harmful or discriminatory to someone online.

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LESSON 3:

Security risks when using digital tools

Learning Objectives

When you have completed this lesson you will be able to explain the main threats to data and computer systems. This includes threats in the form of:

- unauthorised access
- malware such as computer viruses, Trojan horses and spyware attacks
- power failure and surges
- natural disasters
- data corruption
- hardware failure
- network access.

In the previous lesson we looked at the importance of ensuring that when you work with digital tools, you do so in keeping with legislation and relevant guidelines. In this lesson, we focus on some of the security risks you could face when using digital tools. These security risks can have a serious negative impact on the data stored on your computer, as well as the computer system itself. In Lessons 4 and 5 we will go into detail about some of the measures you can put in place to protect yourself from these risks.



Unauthorised access

What do you think is the biggest risk to keeping data safe and secure? Yes, you guessed it – people!

People who are not authorised to access important or sensitive data are a big threat to the integrity of data. People with unauthorised access to important or sensitive data may deliberately corrupt it. They may sell it to others, or they may use it to commit fraud. They may even release sensitive or confidential information to the public. This may affect the organisation's brand, reputation and financial situation. It may also affect you personally, if your personal information is accessed or stolen.

People with unauthorised access to data may be:

Employees	They have the easiest access to computers and data, so form the biggest threats.
Outside users	People with a semi-legitimate reason for accessing data, but who take it too far.
Hackers or Crackers	Hackers gain unauthorised access to a computer system for fun. Crackers do the same thing with the aim of stealing or destroying files.

The easiest way to stop these threats is to restrict access to important or sensitive data. In Lesson 5 we will look at measures that can be used to secure a computer system from unauthorised access.

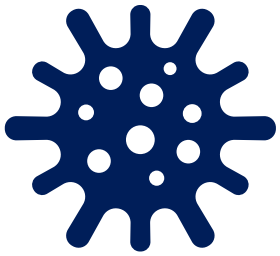
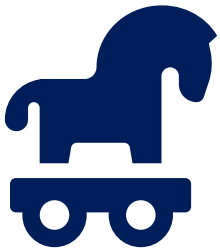
Malware/malicious script

Malware, also known as a malicious script, is dangerous software that may install computer viruses, worms, Trojan horses, ransomware, spyware, adware, scareware, and other malicious programs. It may also track the users' keystrokes to try to discover passwords and credit card numbers.

The following page examines types of malware.

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LESSON 3: **Security risks when using digital tools**

Computer virus 	These are small software programmes that affect the way your computer operates. They are designed to be passed on from computer to computer through email and/or attachments. A computer virus can remain dormant and then suddenly reproduce with great speed by inserting copies of itself into other computer programs, data files, or even the computer's hard drive. A virus will disrupt operation of the computer by deleting, modifying, or corrupting files or software on your computer.
Trojan software 	In computing, a Trojan or Trojan horse is a piece of software hidden within other software that enables others to illegally access your computer or to copy your keystrokes. The main difference between a Trojan horse and a virus is that it does not copy itself once it is on your computer. The most common way for a Trojan to get onto your computer is by you opening an attachment to an email or by running a program you download from a website.
Spyware and adware 	Spyware is computer software that spies on your computer. It acts as an analysis and tracking program, which reports on your preferences and activities without your knowledge. It can also install additional software and redirect browser activity (eg change your homepage). In some cases, spyware can even change computer settings, which can result in slow connection speeds, and also interfere with how the Internet or other programs function, etc. In the next lesson we will look at how virus protection can be used to protect a computer system from all the different forms of malware.
Ransomware 	Ransomware is malicious software that takes over your computer and 'holds it to ransom'. You are locked out of your computer until you agree to pay the ransom. If you don't, the attacker may destroy your files. If you believe you have been affected by a ransomware attack you should contact the police.

LESSON 3: **Security risks when using digital tools**

Try it for Yourself

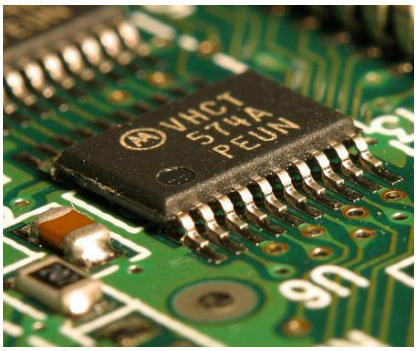
1. From the internet, research an example of a ransomware attack.

2. Use the internet to research the latest malicious software that you need to protect your computer from. Write as many names of the software in the box below.

Computer viruses	
Trojan, Trojan horse	
Spyware	

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Hardware failure



Hardware refers to the electronics and components in a computer. This includes the hard disk or SSD drive, motherboard, and memory. It also includes peripherals such as your monitor, mouse and keyboard.

If your computer hardware fails, it could make it impossible to retrieve the information stored on your computer.

For example, your hard drive could fail as a result of worn out parts or an environmental cause (eg exposure to water, fire, or high magnetic fields). In such cases, you may not even be able to start your computer, let alone access the data on it.

It is important that you:

- keep your computer in good condition
- don't spill liquids on your computer
- report any strange sounds coming from your computer
- move your computer very carefully (if necessary)
- back up all files in case there is a serious hardware failure and the computer is damaged permanently.

Power failures and surges

The power always goes off just when you don't want it to! In a power cut, any work that has not been saved will be lost.

Electricity supply can also surge (or spike) when you turn a computer on or off, or when there is a surge on the electricity grid (eg caused by a lightening strike). Surges in electricity can damage the micro-processors, hard disks, and circuits.

The best way to prevent power failure from affecting your computer is to use an Uninterruptible Power Supply (UPS) and to regularly back up your data. You will learn more about these in Lesson 5.

Natural disasters

Natural disasters include events such as fire, flood, earthquake, or tornado.

Natural disasters are a threat to your computer and to the data or information on it.

- In a flood, the computer may suffer so much water damage that you may not be able to start it at all.
- In an earthquake you may not be able to access your computer because it is buried under rubble.
- In an earthquake or fire, the hard-drive or other important components of the computer may be damaged, making it impossible to start the computer and/or to retrieve data.



Some organisations have disaster recovery plans that guide them through natural disasters so they can get back to 'business as usual' as soon as possible. They may also store a backup copy of their files offsite (ie not in the organisation's buildings) so they do not lose all their data if there is a natural disaster.

When offsite storage of backup is not possible, well-designed buildings, as well as alarm and sprinkler systems can help to limit any damage.

In Lesson 5 we will look in greater detail at how externally stored backups can help protect against data loss as a result of a natural disaster.

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Data corruption

Data that is corrupt is unusable for the purpose intended. The data may be wrong or unreadable. Corruption can happen whenever we are creating, storing, processing or accessing data.

Viruses can cause data corruption. Other common causes of data corruption are:

- hardware fault
- transfer error (eg downloading a file from the internet)
- power failure
- natural disaster.

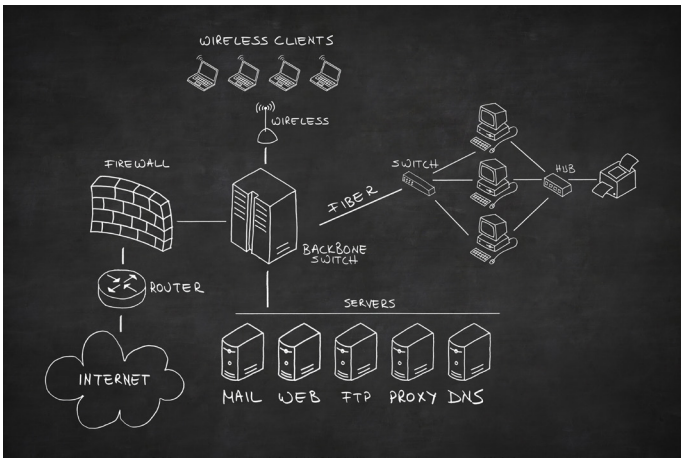
In addition to the measures we have already discussed, important protections that reduce the risk of data corruption are:

- backing up data regularly
- using UPS or surge protectors
- security software to prevent data corruption by viruses, etc.

Network access

A network connection is two or more PCs that either connect through a server or connect directly to each other. They usually share files, devices (eg printers), and even an Internet connection. This is very useful in a work or school environment. It means that instead of each user needing to have their own printer, etc they can share one printer between them.

Unfortunately, the fact that networks make the sharing of resources easier, also means that it makes the system more vulnerable. Having computers on a network can open up opportunities for the spread of viruses, unauthorised access, and harm to the network and/or individual computers. Once one computer on the network is infected with a virus, it can spread quickly to other computers on the network and cause huge damage.



In this lesson, we looked at a range of security risks related to using digital tools. In the following lessons, we look at ways to guard against these risks by implementing basic digital security.

LESSON 4:

Security measures and solutions Part 1

Learning Objectives

When you have completed this lesson you will have an understanding of the following security measures (procedures) and solutions to protect your digital device:

- Anti-malware apps
- Browser settings
- Firewalls

In this lesson and Lesson 5, you will look at the different measures and solutions available to protect your system from data loss due to the risks discussed in Lesson 3. You will start by looking at the software and settings you can use to prevent malware, safe use of your browser and firewalls to prevent downloading malicious software.

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Anti-Malware Apps



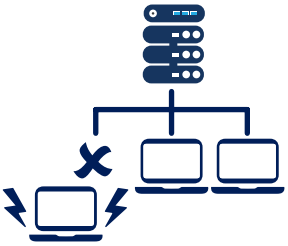




In the previous lesson we looked at the risk malware poses to your computer. In this section we will look at how using anti-malware software can reduce the risk

Anti-malware apps

Using anti-virus or anti-malware software will greatly enhance your chances of avoiding a malware attack.

Protection from malware 	• Install anti-malware software, keep it up to date and scan regularly. • Portable devices such as flash drives should be scanned (using your anti-malware app), before they are used (eg files opened or downloaded from the flash drive). • Do not open suspicious emails or their attachments. • Do not use public computers when you need to enter personal information, eg social media, Internet banking. • Be VERY careful what you download from the Internet. Only download and install software from trusted sources
Symptoms of malware 	• It is sending emails the user did not write. • It crashes and reboots frequently. • Menus and dialogue boxes look strange. • Files are missing or damaged. • It is running slower than usual.
If you think your computer may have malware 	• disconnect it from any network it is on • run your anti-malware software to do a full malware scan • immediately inform the relevant person in your organisation that your device may be infected.

LESSON 4: **Security measures and solutions Part 1**

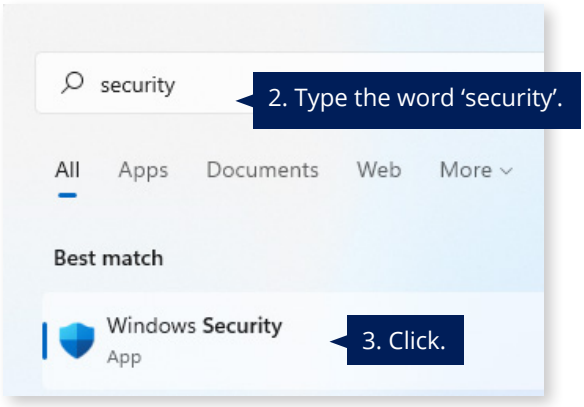
Windows Security App

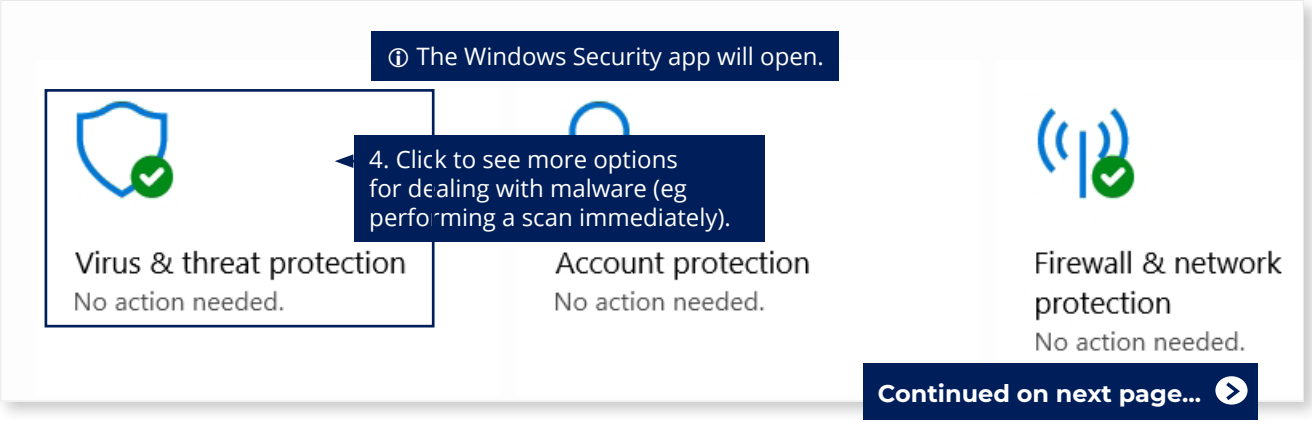
Windows comes with an inbuilt anti-malware program. If you are using a Windows computer, and have no other anti-malware software, Windows Security should automatically be turned on.

The instructions below are for how to check Windows Defender is running, and then use it to scan for malware.

Checking and using Windows Security

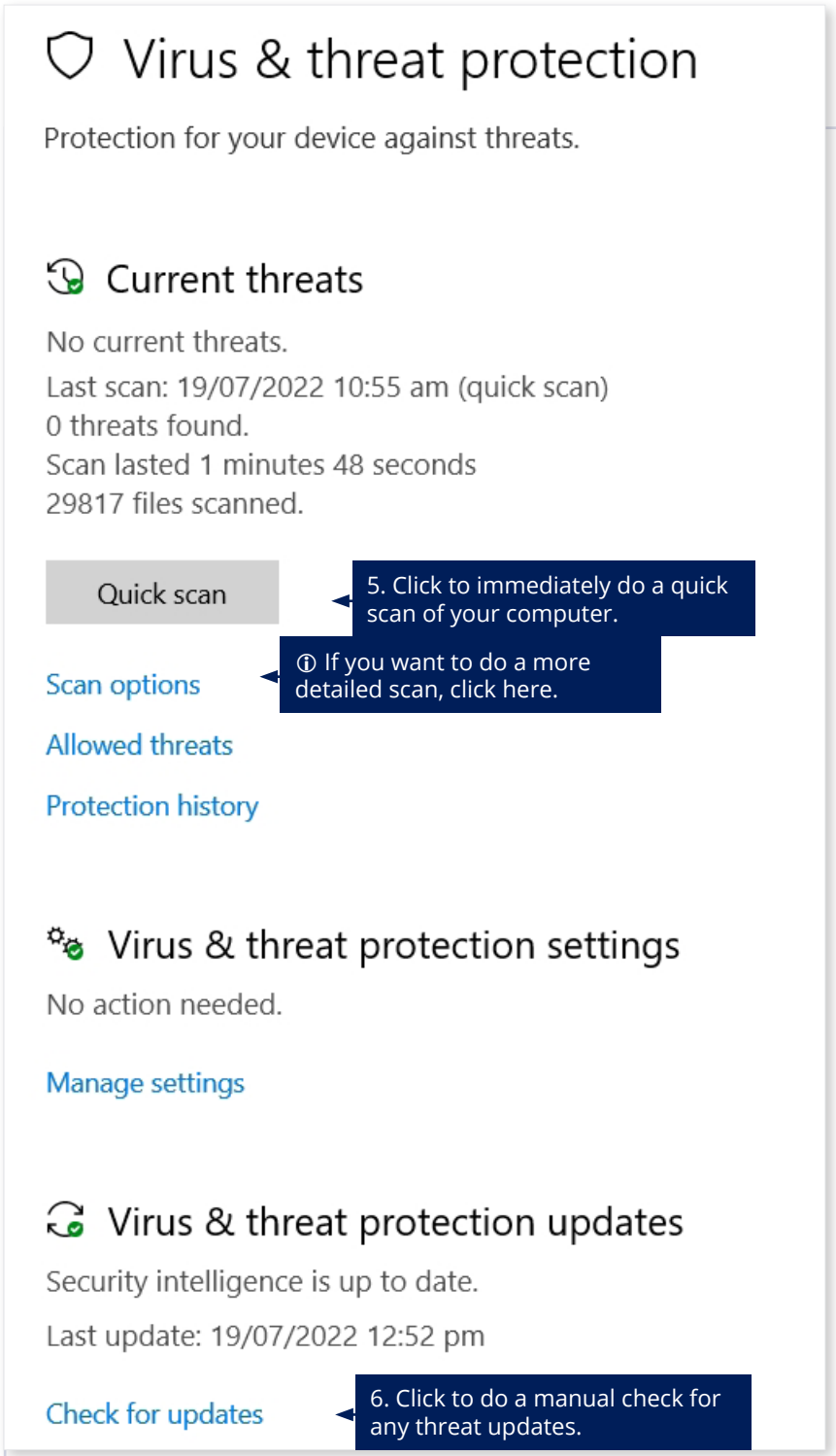
1. Press the ⊞ key on your keyboard.





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Virus & threat protection
Protection for your device against threats.
Current threats
No current threats.
Last scan: 19/07/2022 10:55 am (quick scan)
0 threats found.
Scan lasted 1 minutes 48 seconds
29817 files scanned.
Quick scan
5. Click to immediately do a quick scan of your computer.
Scan options
ⓘ If you want to do a more detailed scan, click here.
Allowed threats
Protection history
Virus & threat protection settings
No action needed.
Manage settings
Virus & threat protection updates
Security intelligence is up to date.
Last update: 19/07/2022 12:52 pm
Check for updates
6. Click to do a manual check for any threat updates.

LESSON 4: **Security measures and solutions Part 1**

The NZ Police have useful information and links on their website (http://www.police.govt.nz) about how to protect yourself and your organisation from malware attacks. Click on the 'Advice' tab, and then select 'Cybercrime and the Internet " or similar. Alternatively, search for 'cybercrime' or 'internet safety' using the search facility on the website.

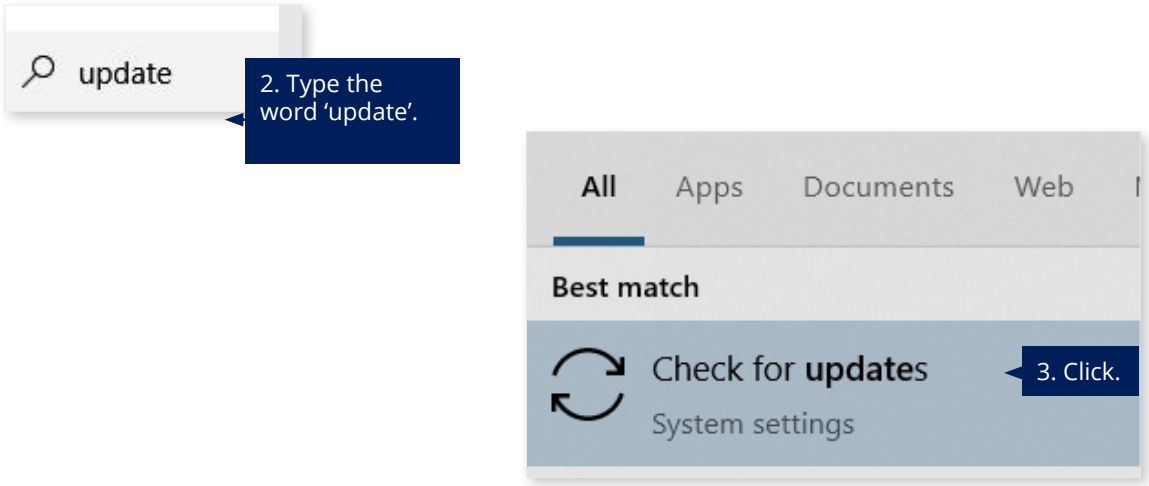
Regular operating system updates

In the same way that regular updates of your anti-malware software can help prevent malware attacks, it is also important to regularly update your operating system (eg Windows, Android, iOS, etc) to ensure the best protection against malware.

This is because all software systems have flaws or vulnerabilities in them. These are ways that hackers and viruses can infiltrate your device. Unethical people are constantly looking for new ways to infiltrate computer systems. To prevent this happening, software manufacturers regularly make updates available which help to fix these security vulnerabilities. These are usually in the form of operating system (OS) patches known as bug fixes.

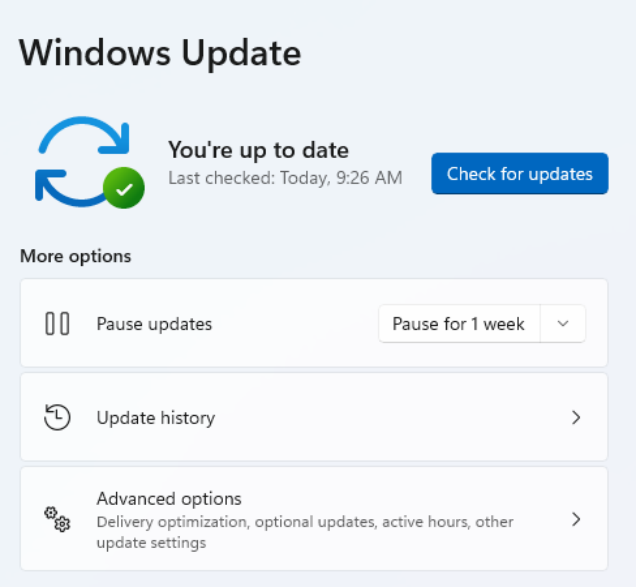
On most devices you will be given a notification of when a new update is available. It is important that when you get such a notification, you update your OS as soon as possible. Usually the notification itself will contain the information you need to perform the update. Otherwise, you can do a manual update via your device settings menu. An example is shown below for updating the Windows OS.

1. Press the ⊞ key on your keyboard.



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LESSON 4: **Security measures and solutions Part 1**



Here you can change your update settings, including when you want automatic updates to occur ('Change active hours' and 'Restart options').



Browser settings

Browsers such as Google Chrome, Firefox, and Microsoft Edge are used to access content on the World Wide Web. As such, they are a potential security risk for your devices. Some organisations have rules regarding which browsers may be used. This is because some browsers are considered to be more secure than others.

Regardless of what browser you use, you need to set your browser's security and privacy settings to help reduce the risk of a data breach or malware attack. In general you should ensure the following has been done for your browser:

- Enable automatic updates for the browser.
- Block pop-ups, plug-ins, and phishing sites.
- **Not** store passwords.
- Disable third-party cookies.
- Enable any other specific browser security and privacy settings your organisation requires.

Try it for Yourself

What are the browser security and privacy settings required by your organisation (ie the place where you are studying or where you work)?

To follow are instructions for how to adjust browser settings for Google Chrome. You would follow similar steps if you are using a different type of browser.

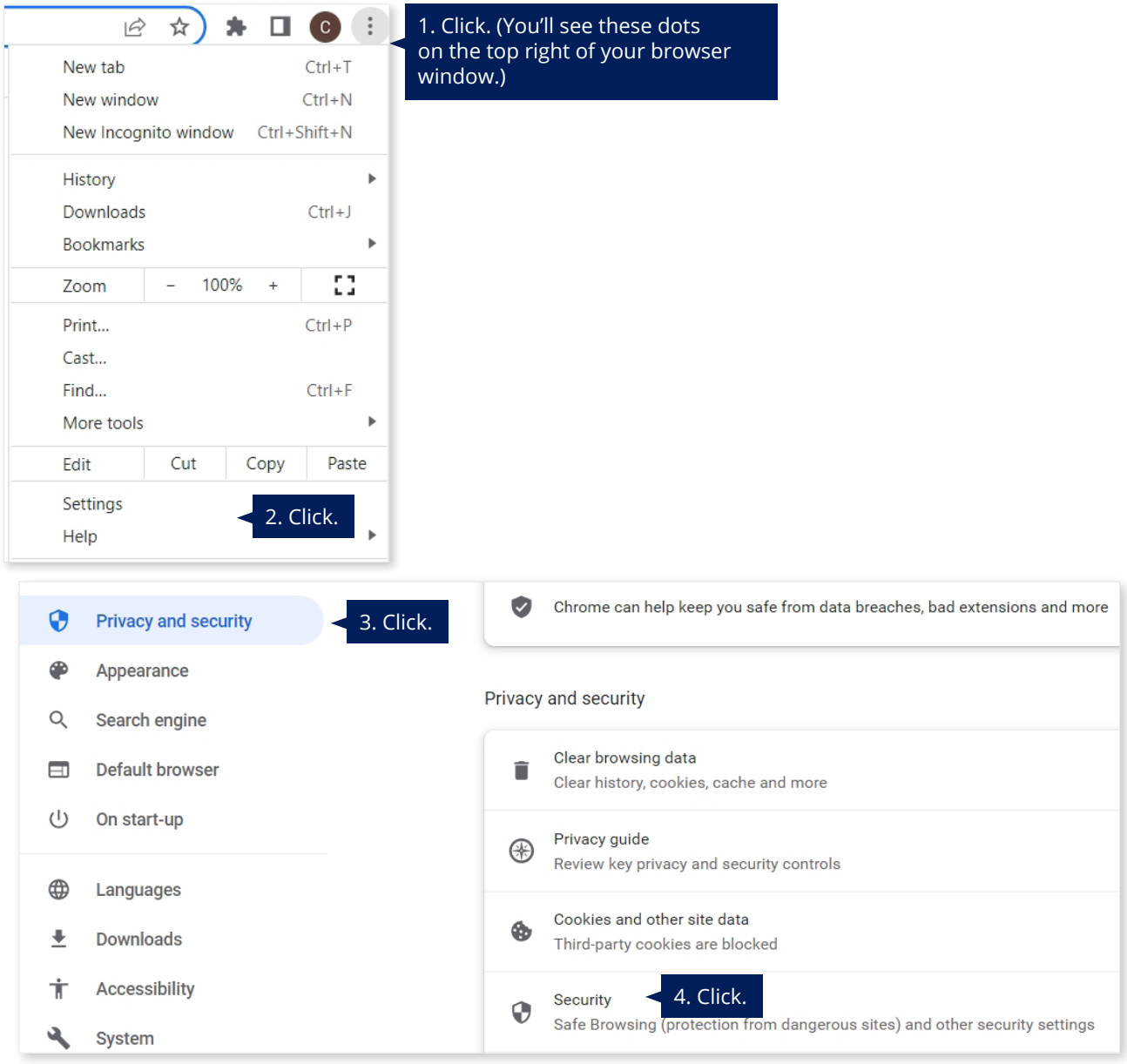
Automatic updates

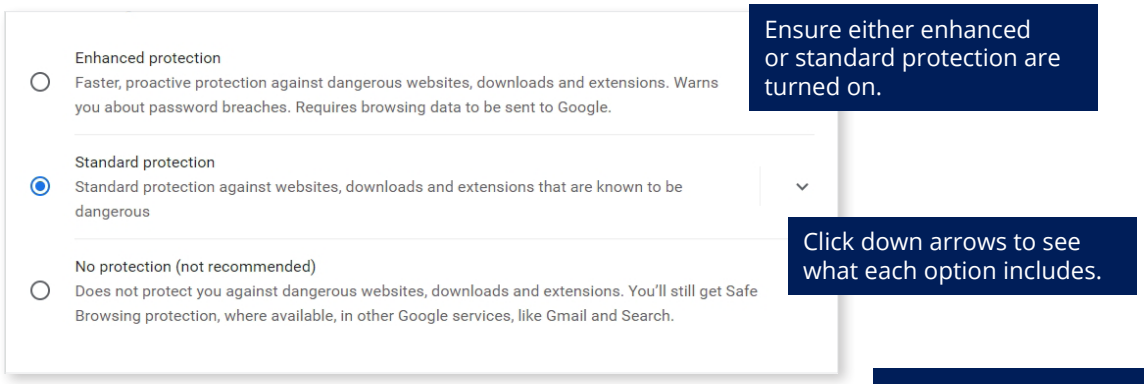
Google Chrome is preset to automatically update. Chrome may occasionally ask you to restart the browser to ensure updates are installed. You will see such a message at the top right of your browser. If you see the message, make sure you close and restart your browser immediately.

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LESSON 4: **Security measures and solutions Part 1**

Google security eg blocking phishing sites



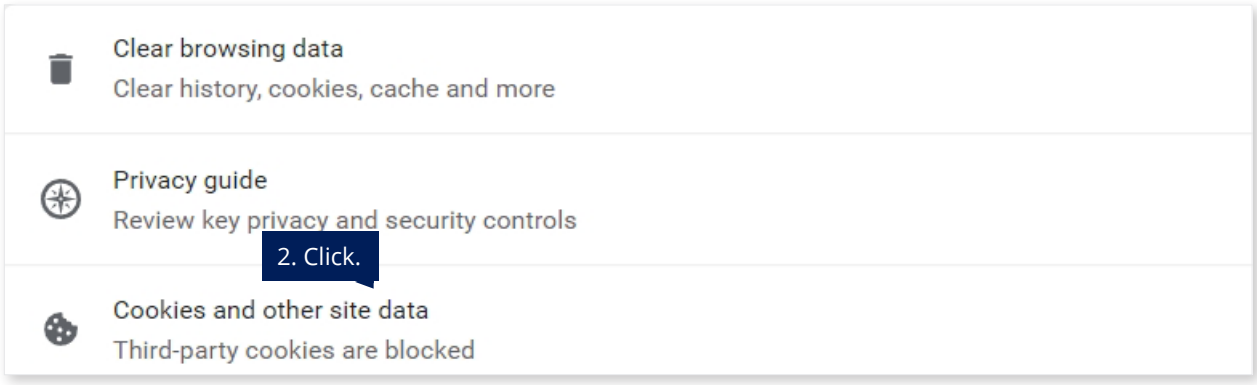


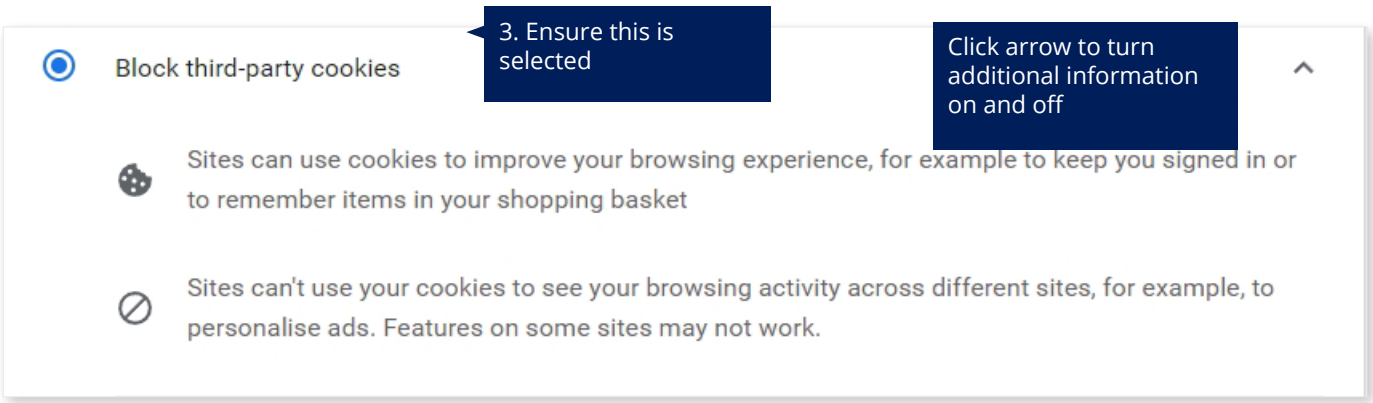
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LESSON 4: **Security measures and solutions Part 1**

Disable third-party cookies

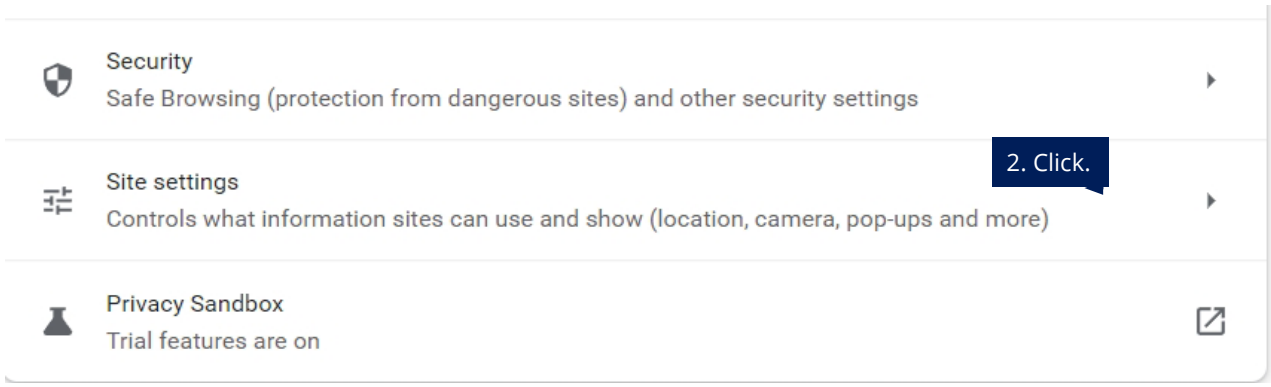
1. Follow steps 1 to 3 on the previous page to access Privacy settings.





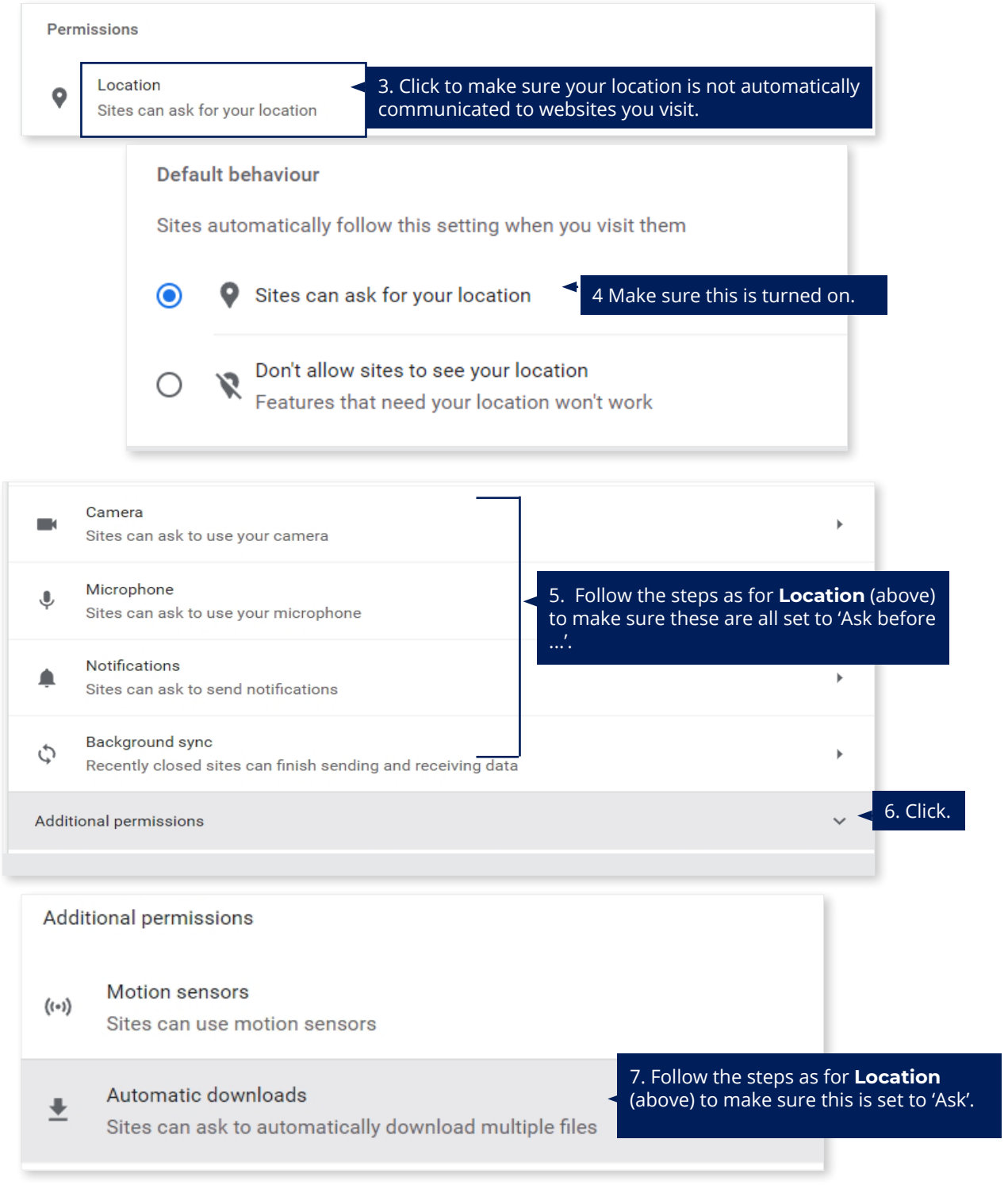
Blocking pop-ups and plug-ins, and other content settings.

1. Follow steps 1-3 on page 49.



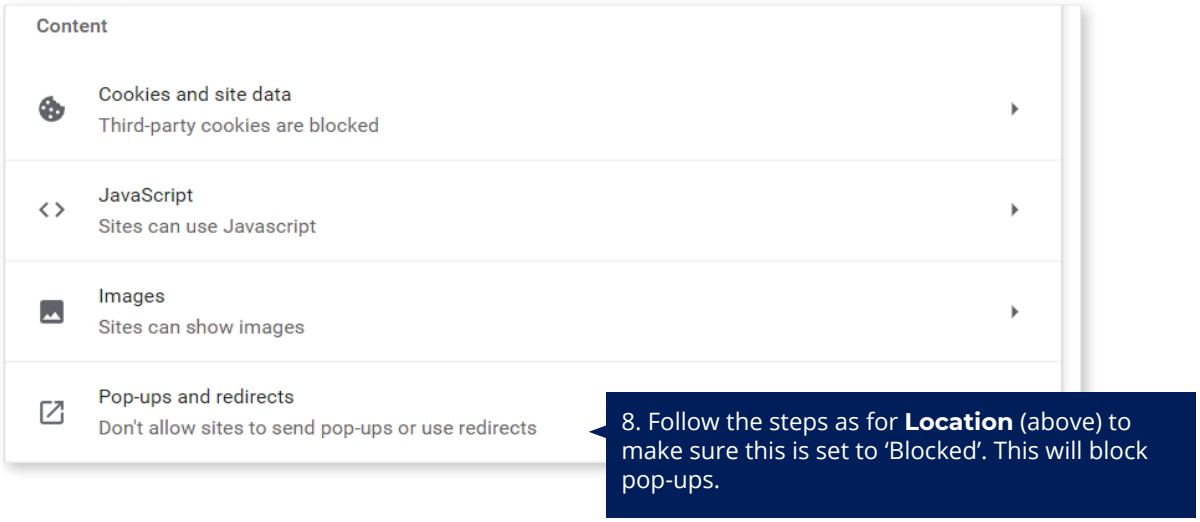
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LESSON 4: **Security measures and solutions Part 1**

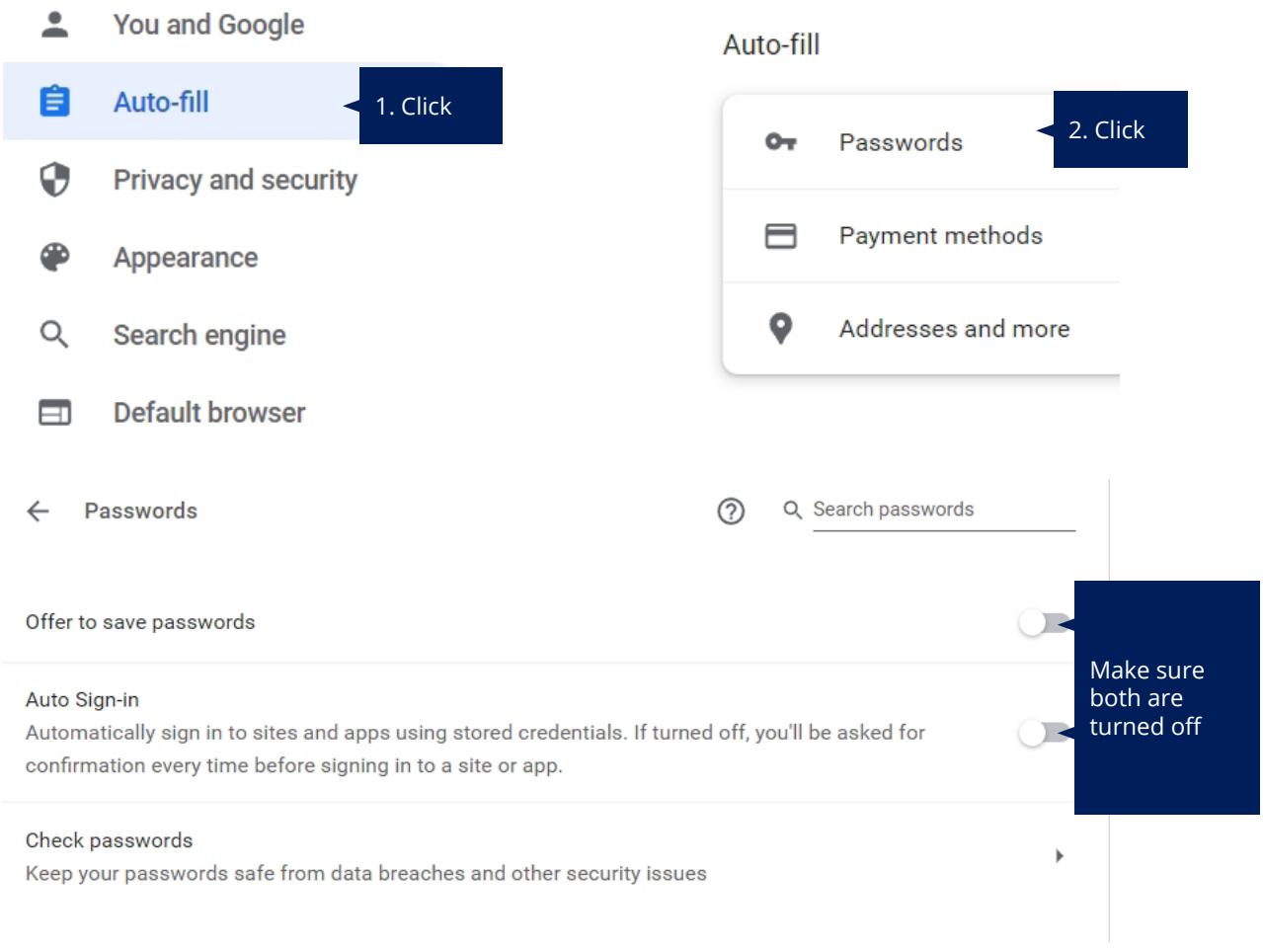


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LESSON 4: **Security measures and solutions Part 1**



To turn off saving passwords



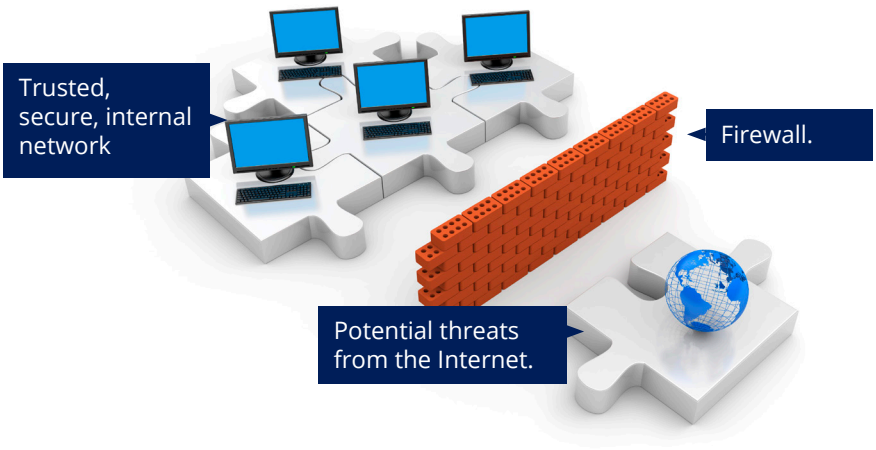
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Personal network protection & firewalls

One of the key ways in which to protect a personal network from threats from the internet, is the use of a firewall.

A firewall works by restricting or preventing the spread of computer viruses and Trojans.

- It controls the incoming and outgoing data from the internet based on a particular set of rules.

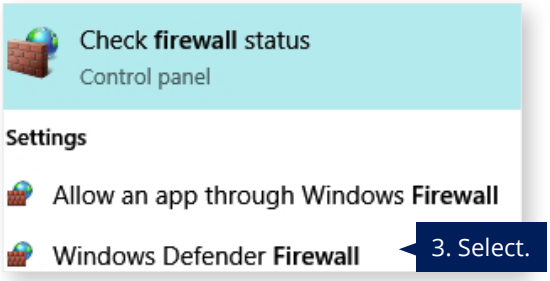


- It establishes a 'wall' between a trusted, secure internal network – your workplace or schools computer system – and another network that cannot be trusted (ie the Internet).

Below are instructions for how to access the inbuilt Windows firewall settings.

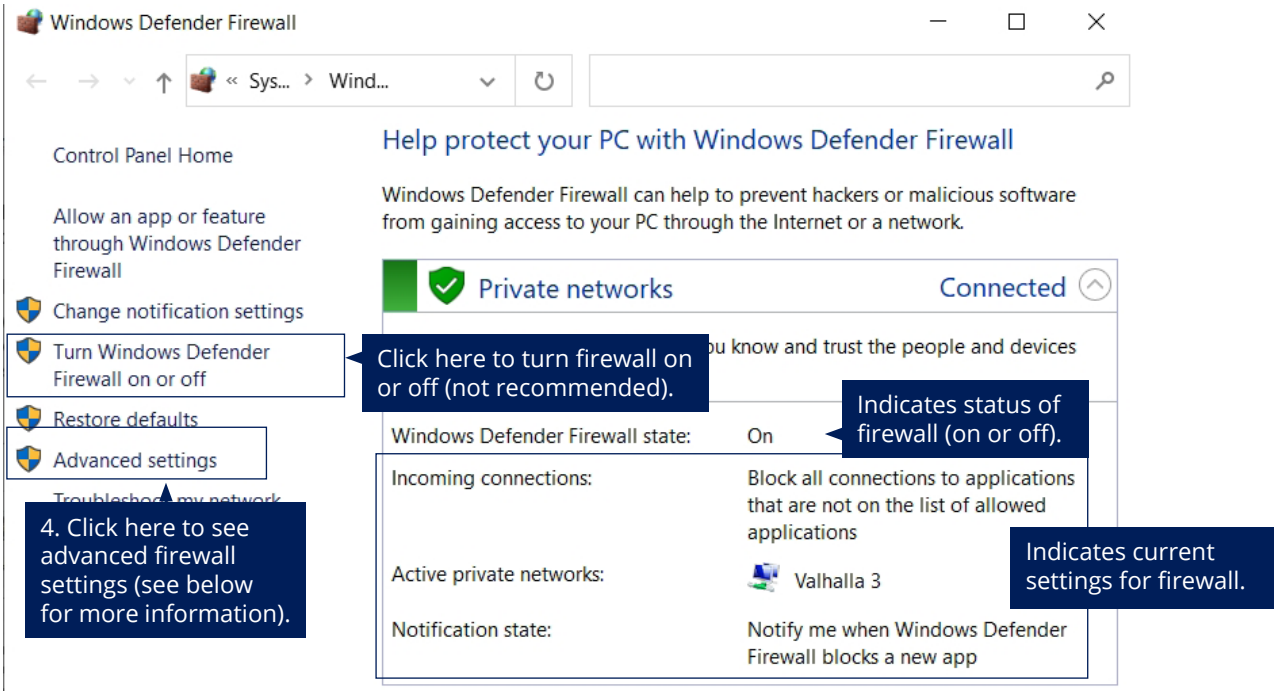
1. Press the ⊞ key on your keyboard.





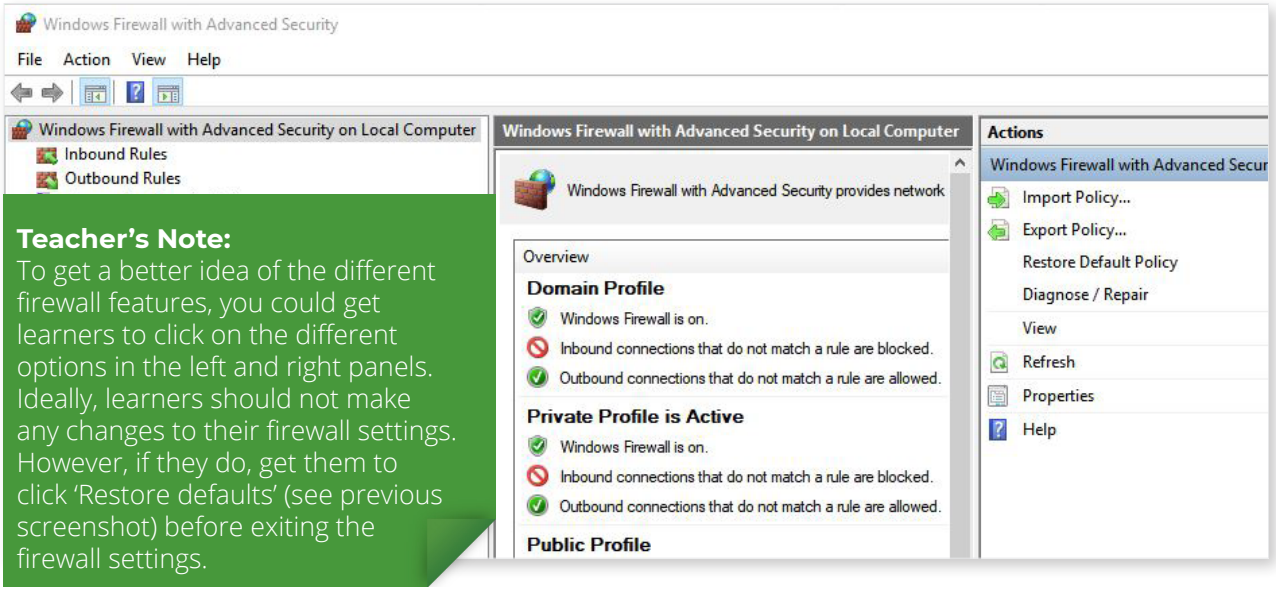
LESSON 4: **Security measures and solutions Part 1**

The Windows Defender Firewall settings (in Control Panel) will open (see below).



When you click on 'Advanced settings', a window like the one below will open. Here you can see more detailed information about the firewall settings (centre panel). You can also adjust and set inbound, outbound, and connection security rules (left panel).

NOTE: In most contexts, the default Windows firewall settings are appropriate. You should not make changes to these firewall settings unless you have been specifically instructed to do so by your organisation.



Teacher's Note:
To get a better idea of the different firewall features, you could get learners to click on the different options in the left and right panels. Ideally, learners should not make any changes to their firewall settings. However, if they do, get them to click 'Restore defaults' (see previous screenshot) before exiting the firewall settings.

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LESSON 5:

Security measures and solutions Part 2

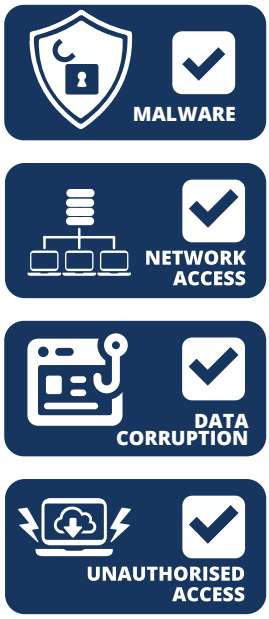
Learning Objectives

When you have completed this lesson you will be able to identify and describe the following security measures (procedures) and solutions to ensure security when using digital tools.

- Access control measures (eg passwords, read only files, etc)
- Surge protection
- Backup measures (eg saving files, etc)
- Maintaining security measures

In the previous lesson we looked at some ways to protect your system. In this lesson, we will look at further measures and solutions you can put in place to help reduce the risk of unauthorised access and damage to your hardware. You will also learn about the importance of maintaining the security measures you put in place.

Access control



People who are not authorised to access important or sensitive data are a big threat to the integrity of data. This ranges from the theft of confidential information, to the intentional damage to data, or the system.

We are going to look at some ways end users can help to prevent unauthorised access to digital devices and the computer systems in an organisation. In particular, you will learn about:

- passwords and other tools to restrict access
- logging-off and shutting down
- read-only files
- physical security measures.

Passwords and other tools to restrict access

It is important that access to all devices is restricted using appropriate tools.

Many newer devices offer a range of options for securing digital devices. These could include using a PIN, pattern lock, fingerprint, eye scanner lock, voice recognition, two-factor authorisation or a password.

The type of security to be used will depend on what is available on the device and what the end user finds most convenient. But what is most important is that the end user ensures they use some form of security.



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LESSON 5: **Security measures and solutions Part 2**

If a password is being used, it needs to be a unique, strong password.

- **Unique:** Don't use the same password for your device that you use for your accounts or applications. Using the same password makes it easier for a hacker to get into more places.
- **Strong:** Guidelines for creating strong passwords are as follows:
    - Include a mix of numbers, punctuation, and upper and lower-case letters.
    - Preferably use a phrase (group of words), or a password that is at least eight characters long.
    - Avoid using repetition, dictionary words, sequences of letters or numbers, usernames, or personal information such as names or dates.

The way that you access the security settings on your device will depend on the type of device you are using.

Two-factor authentication (2FA)

This offers extra security by using a second step to provide something as well as a password in order to access a system or account.

Examples are:

- security questions
- a code that is sent to your mobile phone or other device
- a code from an app such as Google Authenticator



LESSON 5: **Security measures and solutions Part 2**



Try it for Yourself

Answer the following questions in relation to the organisation that you work for (or your place of learning).

1. Are all users at your organisation required to have a username and password?

2. How often do passwords have to be changed?

3. Are there any rules about sharing passwords with others? If so, list these rules.

Teacher's Note:

Answers will be specific to the learner.

Password protect a file

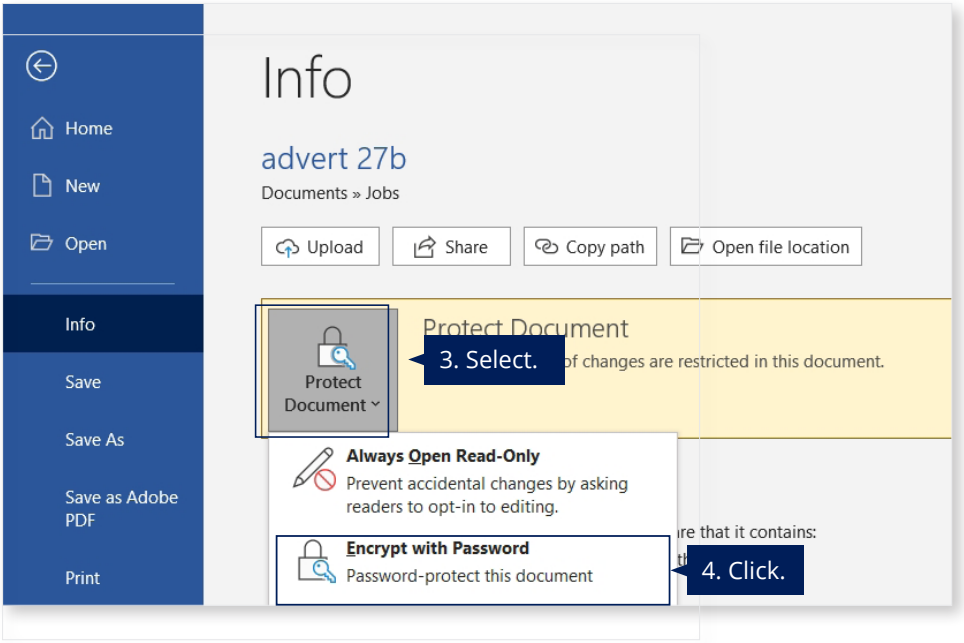
In addition to adding a password to your device, you can also add password protection to individual files. This will mean that anyone who wants to open the file needs to first enter a password.

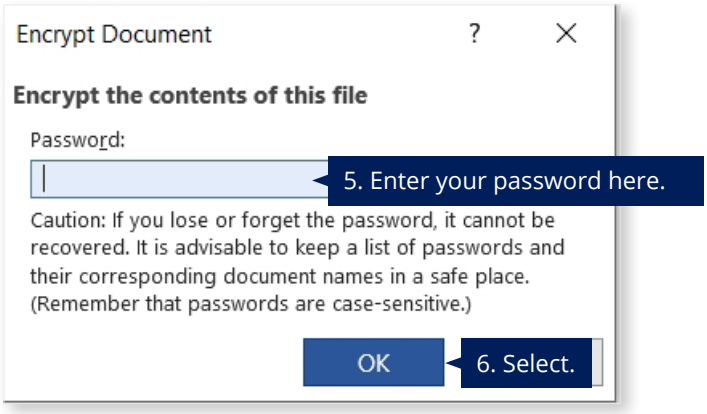
An example follows for how to apply password protection to a Word document. The process is similar for other types of files.

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LESSON 5: **Security measures and solutions Part 2**

1. Open the file to which you want to apply permissions.
2. Select the **File** tab on the Ribbon.





7. You will be asked to enter your password a second time.
8. Save the file for the change to take effect.

Once a password has been set and you close the file, when you try to open it again, you will need to enter a password.

LESSON 5: **Security measures and solutions Part 2**

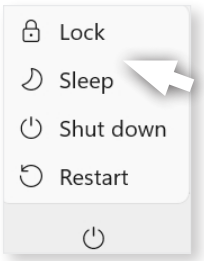
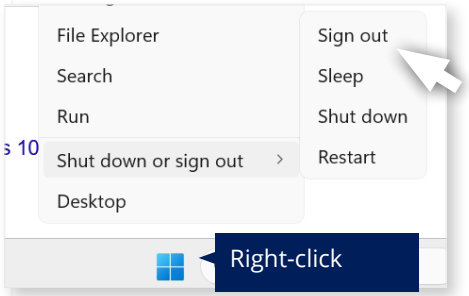
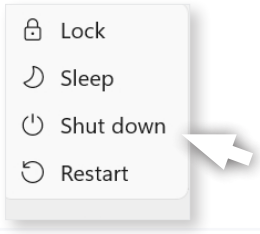
Secure systems

A secure computer system will have a password. You will notice that Windows requires a password or pin number to gain access to your desktop and files.

These are used to prevent someone physically gaining access to your files when you are not at your computer. When you step away from your computer, you should either:

- Activate the lock screen
- Sign out
- Shut down.

The following explains the purpose of each.

Lock From Start Menu 	When you are away from your computer for a short time, you should 'lock' access to it. This keeps your programs running but no-one can access your desktop without entering a password.
Sign out 	This can also be referred to as log out. This signs you out of the system (or your desktop for Windows 10). The computer is still switched on but your profile is closed and cannot be accessed without entering a username or password. Amy programs running will be closed down. This can also be useful if someone else needs to use the computer – they can sign in with their own username and password. Note: If the account is set up in Windows, it will not ask for the user name, just the password. Your organisation may require computers to be signed out when leaving for extended periods such as lunch.
Shut down From Start Menu 	Shutting down a computer refers to turning it off. Some organisations require all computers to be shut down at the end of the day. Your username and/or password will need to be entered when it is next turned on.

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LESSON 5: **Security measures and solutions Part 2**

To lock Windows

1. Press the Windows key + L on your keyboard.

This will bring up your wallpaper and you will need to enter your PIN or password to get back into your desktop.



To sign out or shut down

Press Alt+F4 and select your option from the dropdown list.

Try it for Yourself

In the boxes below, write when you might lock your computer and when you would log off.

Lock a computer	*Going to the bathroom* *Going to lunch* *Collecting printing from the printer* *Going to a meeting*
Signing out	*At the end of the day* *If I am not using the computer anymore* *If someone else would like to use the computer and has a different profile*

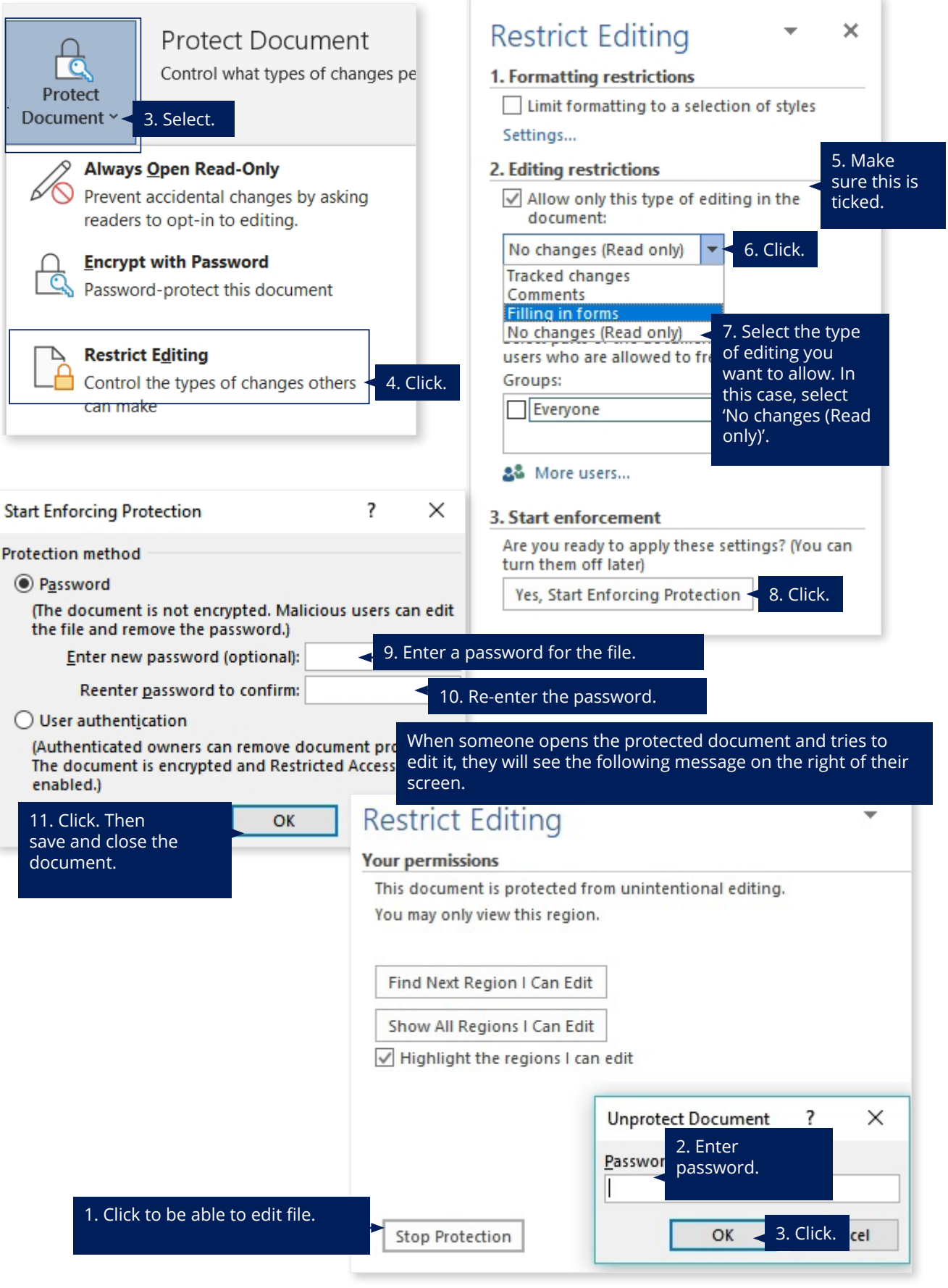
Read-only files

You have already seen how to password protect a file. It is also possible to set other access controls on individual files. For example, you can restrict what people who access the file are able to do. You can make the file 'read-only', which means that people can open the file and read what is in it, but they cannot make any changes to the file. To follow is an example of how to apply access permissions to a file, using Microsoft Word. The process is similar for other file types.

1. Open the file you want to make read-only.
2. Press the **File** tab on the Ribbon.

Continued on next page...

LESSON 5: **Security measures and solutions Part 2**



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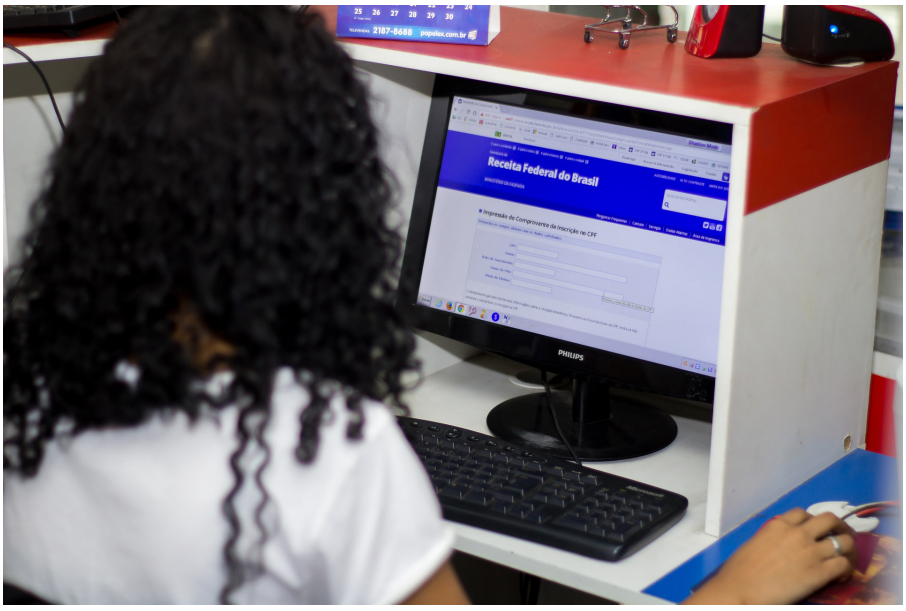
LESSON 5: **Security measures and solutions Part 2**

Physical security

The fewer people who have access to the place where important and sensitive data is stored, the less likely that someone will use the data in a way that they shouldn't. Making sure that only authorised people have physical access to data, will help protect its integrity.

Ways organisations may do this include:

- Swipecard access to non-public areas
- Physically locking away devices when they are not there
- Requirements regarding storage of portable devices when away from the office
- Always have screens facing away from visitors so they cannot read what is displayed on them.



LESSON 5: **Security measures and solutions Part 2**

Another key way to protect your personal network is to set access permissions on your internet connection. For example, this could include the following:

- Setting a password for your home network, especially if you have a wireless network. This will mean that people close to your home are not able to access and use your wireless connection to the internet.
- Setting a password on your router. Many routers have default login creditials (eg using a password like 'admin'). As soon as you get a new router, make sure you set a new password that someone else will not be able to easily guess.

Surge protector and UPS



While anti-malware software, browser settings, and firewalls can help protect your computer from malware and unauthorised access, a surge protector and a UPS can help prevent physical damage to your computer from a power surge.



Surge protectors (or surge suppressors) provide some protection against sudden changes in the power supply that could cause your computer to crash. An electrical surge is a temporary, very short (less than a second) increase in the electrical supply voltage (or current, or both). The surge protector works by blocking or by shorting to ground any unwanted voltages above a safe threshold.



A **UPS (Uninterrupted Power Supply or Uninterruptible Power Supply)** gives your computer a continuous power supply. If the power is disrupted, it gives the user a few minutes time to save their files and shut down their computer.

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Backup measures



Making sure you have backup measures in place can help ensure that even if there is a power failure, natural disaster, or other event that causes your computer to shut down unexpectedly, you will still have access to your data. In this section we look at measures you can take to save and backup your data. We will also look at how you can have a backup power supply, to allow you to keep working on your computer even if there is a power failiure.

Saving, backing-up, and restoring data

One of the best ways to ensure the integrity of data is to save data regularly. If an error occurs with a computer (such as getting infected with a virus, or a hard drive failure), the computer can be restored to the point before it got infected or the data was lost.

Saving data

Good practice is to save your work as you go. This can be done by setting your computer to autosave regularly. Using autosave regularly will mean that, if your computer crashes, you may only lose a small amount of work. The more regularly you save your work, the less data you are likely to lose.

Backing up data

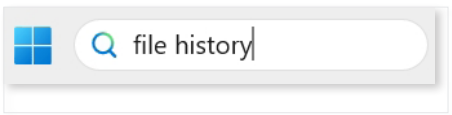
Backing up your data to a separate location is also good practice.

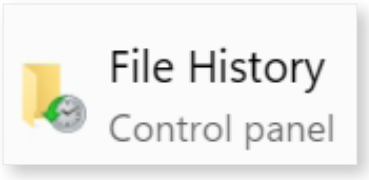
Backup using File History backs up files that are in your This PC directory, contacts, favourites, and on your desktop to a separate location. If the originals are lost, damaged, or deleted, you can restore all of them.

To turn File History on:

1. Press  on your keyboard and start typing the words 'File history'.

As you start typing, the search window will open.

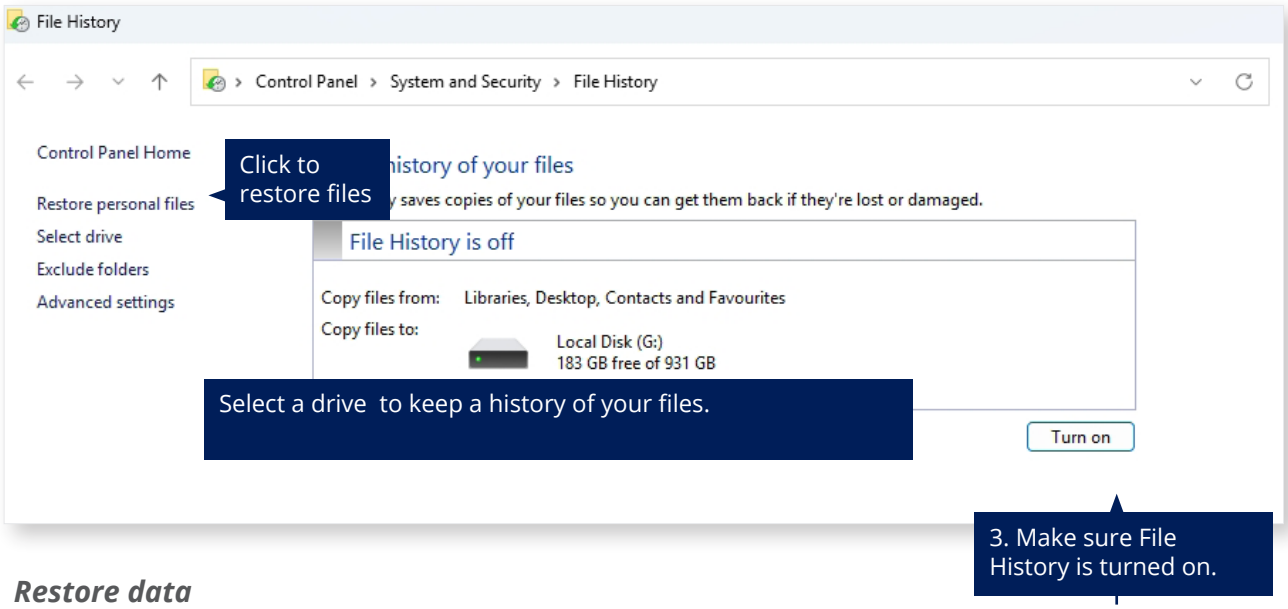




2. Select.

Continued on next page...

LESSON 5: **Security measures and solutions Part 2**



Restore data

Using File History allows you to easily restore data from the backup location. You can even find different, earlier versions of your files.

Organisation backup

Most organisations, including workplaces and schools, will have their own management plans for backing-up and restoring data. These may outline different types of backup, and disaster recovery plans.

These plans may also include the storage of backup data by an online backup service. An online backup service allows important data to be backed up over the Internet and securely stored on a server in a professional data centre offsite.

An organisation could also back up its data to the 'cloud'. You too can backup your own data using such cloud services as Google Drive, One Drive and Dropbox.

The advantage of having your important data backed up offsite, away from your home or office, is that it's safe from theft, fire, and other local disasters.

Teacher's Note:

Steps for how to perform local and external (Cloud-based) backups is covered in depth in the **Instant Education Solutions** resource for US29772v2 (Manage files and folders using digital devices).



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LESSON 5: **Security measures and solutions Part 2**



Try it for Yourself

What does your place of learning or workplace do to back up its data? Does it back up onsite, offsite or to the cloud?

Teacher's Note:

Answers will be specific to the learner.

What does your place of learning/workplace require you to do in terms of saving your work and backing up your data? If there are no written policies regarding saving and backing up your work, check with your manager or teacher.

Teacher's Note:

It is important that the learner is aware of their organisation's backup plan and policies. For example, it may be against policy to copy data onto remote devices.

Maintaining security measures

In this lesson you have learned about procedures for managing risks to your devices. Once you have implemented these measures, they need to be maintained. Examples of maintaining your security measures include:

- **Changing your password or PIN regularly**

If someone has accessed your password, you may not know. Changing it regularly helps prevent this.

- **Keeping your software up to date.**

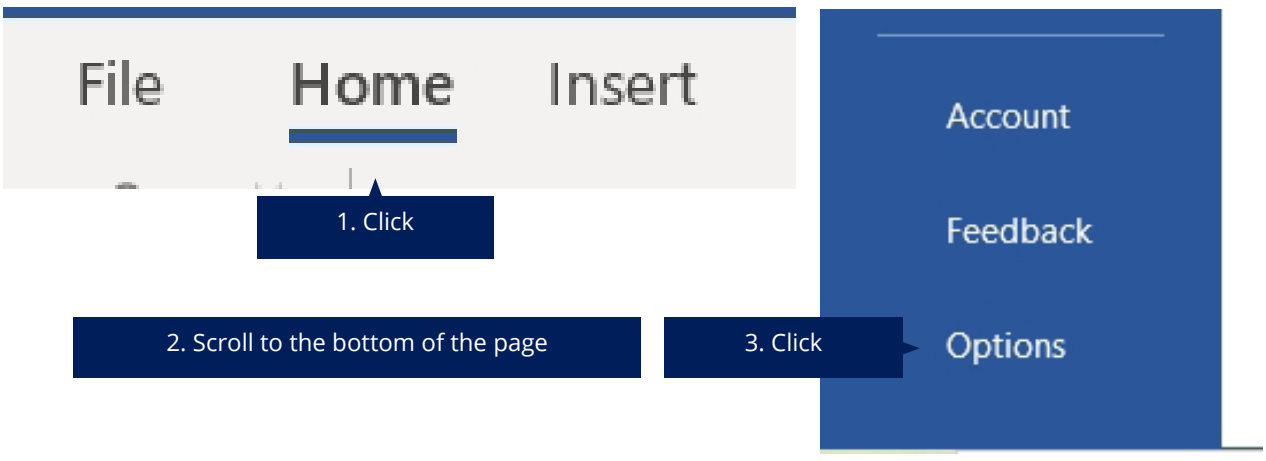
It is vital that anti-malware software is kept up to date. New malware is being created all the time, so the anti-malware software's database needs updating regularly.

Operating system and application software updates are often to fix security issues. It is important to make sure these are regularly updated. This applies to all devices including phones, tablets, and computers.

- **Save regularly**

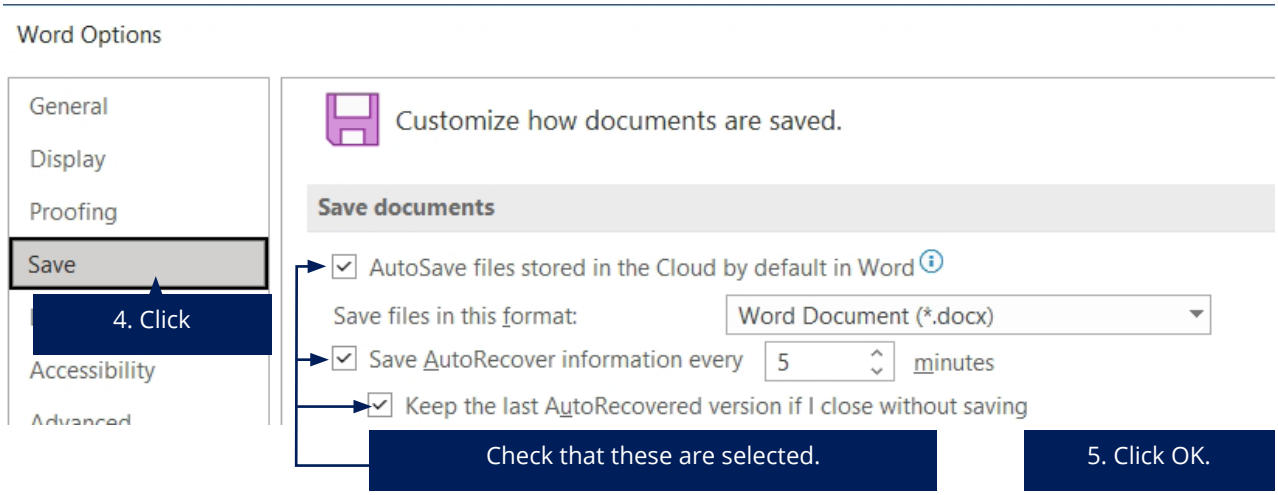
As discussed in the section on backups, saving your work regularly can help prevent issues with sudden power loss or issues with your software. You can do this using keyboard shortcuts (usually Ctrl + S) at regular intervals, or by using autosave.

To make sure autosave is turned on (Using Word as an example):



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LESSON 5: **Security measures and solutions Part 2**



- **Check backups are working**

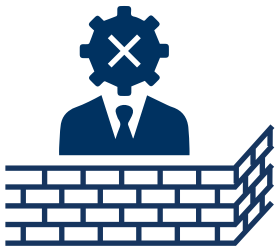
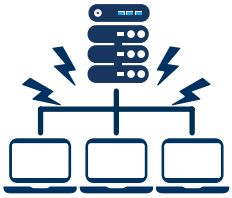
Regularly check that your backup system is working correctly.

- **Log out and shut down**

Make sure that others cannot access your system when you are away from it.

LESSON 5: **Security measures and solutions Part 2**

Lesson 3 Review Sheet

RISK	SOLUTION	MAINTENANCE
Unauthorised access 	Log-off Shut-down Firewall Browser settings Passwords for device and router/ISP Physical security Controlling access	Change regularly
Malware 	Firewall Anti-malware software	Update regularly
Natural disaster 	Off-site backup	Backup regularly
Data corruption 	Backup Read-only files Anti-malware software Save frequently	Backup regularly Update regularly
Hardware failure 	UPS or surge protector	
Network access 	Log off Use passwords	Update regularly

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LESSON 5: **Security measures and solutions Part 2**

Try it for Yourself

In the workplace or learning environment, some security functions are controlled by the information technology department. However, there are still plenty of security measures you can take. There are also measures you can take to keep your personal devices safe.

List all the security measures you can take in your workplace/place of study and personal devices to keep your data safe. Make sure you know how to implement these

Workplace/place of learning

- *Log-off and shut down my computer when I'm not using it.*
- *Change my password regularly.*
- *Be careful of using USB sticks to share files, use the cloud instead.*
- *Keep my work devices secure – do not leave them in public areas.*
- *Make documents or files I don't want changed, read only.*

Teacher's Note:

Answers will depend on the learner's workplace or training provider's policies and procedures.

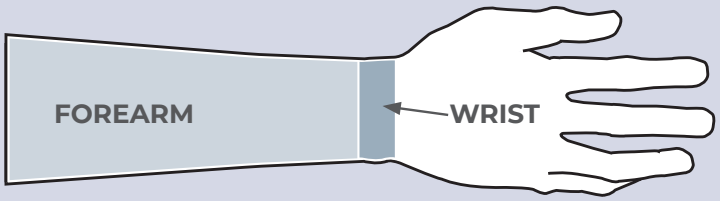
Personal devices

- *Back up my phone and laptop*
- *Make sure my firewall is turned on*
- *Install and use anti-malware software, and make sure it is updated.*
- *Log out of apps and shut my computer down when not in use*
- *Do not save passwords.*
- *Make sure my browser's security settings are enabled.*
- *Change my passwords regularly, including those for my router and ISP.*
- *Use a UPS protector for my devices.*

Teacher's Note:

It is important that the learner is aware Answers will depend on the learner's devices and their access to WiFi and the internet.

Glossary

airborne pollution	something poisonous or dangerous (pollution) that is carried in/by the air (airborne)
adjusted	can be changed
alter	change
consent	agree to let someone do something
contrast	difference
cyberstalking	following someone's movements online for the purposes of annoying, bothering or intimidating them
data breach	when an unauthorised person gets access to personal information, either intentionally or by mistake
deliberate(ly)	on purpose (not by mistake/accident)
digital environment	a situation or place where digital tools are used; this includes both hardware (digital devices, including storage and display device), software (applications), and data in digital form
discomfort	not feeling comfortable, feeling an ache, tenderness, etc
distribute	give to others; share
dormant	asleep; inactive
ergonomic	the arrangement and use of furniture and digital device components in a way that helps to prevent and manage discomfort, pain, and injury
ethical	following morals and principles - displaying respectful and responsible behaviour.
fatigue	tiredness
firewall	software used to protect a computer system against threats from the internet
forearm	
glare	bright light that makes it difficult to see
implement	put into practice
infiltrate	get access to something
injury	a physical wound or damage to the body, sometimes needing medical attention

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integrity	for data to have integrity it should not have been changed, lost or destroyed
intentional	done on purpose (**not** by mistake or accident)
keystrokes	the keys you press on your keyboard (ie what you type)
legitimate	genuine; real 'Legitimate computer programs' refers to software intentionally installed on a computer that is performing an expected task/function.
malicious	intended to do harm
malware	short for 'malicious software'; software that is intended to do harm (eg a virus)
organisation	business, place of learning, etc
personal information	anything that can identify a person (eg a photograph of the person, their name, address, etc)
phishing	phishing (pronounced 'fishing') is a way people try to get sensitive information such as your username, password, credit card details, etc
postures	how you position your body
psychological	to do with how you feel emotionally and mentally
repetitive	again and again
spam	an electronic message you did not ask for
unauthorised	not given permission An unauthorised person is someone who has not been given permission to access or do something.
unintentional(ly)	not on purpose; by mistake
unsolicited	not asked for/requested
ventilated	getting air to come into a room/building and move around the area freely
vulnerabilities	weaknesses; at risk of being attacked or harmed

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